IEC Low Voltage Induction Motors 400 V 50 Hz

Motors for all applications

Making you more competitive

ABB has been manufacturing motors for over 100 years. Our products are designed to be reliable, efficient and cost effective, and we can supply motors for practically any application. A full range of services is available through our worldwide service organization, with the latest eBusiness systems providing round-the-clock access, easy ordering and fast delivery.

ABB (www.abb.com) is a leader in power and automation technologies that enable utility and industry customers to improve performance while lowering environmental impact. The ABB Group of companies operates in around 100 countries and employs about 108,000 people.

General Purpose Motors

ABB's General purpose motors are readily available from central stock locations and distributors throughout the world. While designed for standard and straightforward uses, the motors can be modified to meet most specifications. Built to the highest manufacturing standards, the General purpose motors use the best materials sourced from around the world. This brings a quality and reliability that can see motors operating for over 30 years. Competitively priced, the motors meet EFF2 energy efficient classification, with EFF1 as option.

Process Performance Motors

ABB's Process performance motor is engineered to meet the most demanding applications found in industries including pulp and paper, water treatment, food and beverage, metals and building materials. Such is the high design specification of the motor, when used in conjunction with applications in these industries, ABB is able to provide a 3-year warranty.

Built to the highest manufacturing standards, the process performance motors use the best materials sourced from around the world. This brings a quality and reliability that can see motors operating for over 30 years. Competitively priced, the motors meet EFF1 energy efficient classification.

Availability

Through our extended support and services such as an efficient global stock concept, we provide you with easy ordering and quick delivery.

– EDI based ordering
– Over 300,000 motors stocked worldwide
– Most frequently used OEM motors stocked in depth
– 2 to 72 hours delivery for stocked motors
– Modifications within 24 hours

BusinessOnline

BusinessOnline, at http//online.abb.com/bol gives you real-time, on-line access to your own personal portal to ABB motors and drives. You can choose, configure and order products, determine their availability and stock levels, follow their progress through the order-processing and delivery chains, and access a wealth of support services and technical information such as drawings, test results and technical documentation.

IEC Low Voltage Induction Motors
400 V 50 Hz

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ABB reserves the right to change
the design, technical specification and
dimensions without prior notice.

Detailed product information

This combination catalogue is intended to give brief summary on technical data and dimensions for our General purpose and Process performance motors. More detailed information on the ranges can be found from following product catalogues:

General Purpose Motors EN
Process Performance Motors EN

Contact your local Sales office for the catalogue or download if from our web site www.abb.com/motors&drives.

Ordering information

Sample order

When placing an order, please state the following minimum data in the order, as in the example.

The product code of the motor is composed in accordance with the following example.

Motor type	M3AA 112 MA
Pole number	4
Mounting arrangement (IM-code)	IM B3 (IM 1001)
Rated output	3 kW
Product code	3GAA 112021-ADA
Variant codes if needed	

Motor size

A	B	C	D, E, F
M3AA	112 MA	3GAA 112 021 - ADE, 003, ...	
		1 2 3 4 5 6 7 8 9 10 11 12 13 14	

A Motor type
B Motor size
C Product code
D Code for mounting arrangement
E Voltage and frequency code
F Generation code followed by variant codes

Explanation of the product code

Positions 1 and 2
3G = Business area LV Motors

Position 3-4
Enclosure and stator frame material
3GVA, 3GAA, 3GAP = Totally enclosed motor with aluminum stator frame
3GQA, 3GBA, 3GBP = Totally enclosed motor with cast iron stator frame
3GCA = Totally enclosed motor with steel stator frame

Position 4
Type of rotor
A = Squirrel cage
P = Process performance motor

Positions 5 and 6
IEC size

05 = 56	**11** = 112	**25** = 250
06 = 63	**13** = 132	**28** = 280
07 = 71	**16** = 160	**31** = 315
08 = 80	**18** = 180	**35** = 355
09 = 90	**20** = 200	**40** = 400
10 = 100	**22** = 225	**45** = 450

Position 7
Pole pairs
1 = 2 poles
2 = 4 poles
3 = 6 poles
4 = 8 poles
5 = 10 poles
6 = 12 poles
7 = > 12 poles
8 = Two-speed motors
9 = Multi-speed motors

Positions 8 to 10
Running number

Position 11
- (dash)

Position 12
Mounting arrangement
A = Foot-mounted motor, top mounted terminal box.
B = Flange-mounted motor. Large flange.
C = Flange-mounted motor. Small flange.
F = Foot- and flange-mounted motor. Special flange.
H = Foot- and flange-mounted motor. Large flange with clearance holes.
J = Foot- and flange-mounted motor. Small flange with tapped holes.
L = Foot-mounted, terminal box on LHS seen from D-end.
N = Flange-mounted (CI ring flange FF)
P = Foot-and flange-mounted motor (CI ring flange FF)
R = Foot-mounted, terminal box on RHS seen from D-end.
S = Foot- and flange-mounted, terminal box RHS seen from D-end.
T = Foot- and flange-mounted, terminal box LHS seen from D-end.
V = Flange-mounted motor. Special flange.

Position 13
Voltage and frequency: See tables below

Position 14
Version **A,B,C...** =
Generation code followed by variant codes

Single speed motors	Code	Single speed motors	Code
380 VY 50 Hz	**A**	200 VD 60 HZ	**P**
380 VD 50 Hz	**B**	440 VY 50 Hz / 480 VY 60 Hz	**Q**
400 VD 50 Hz (mid range value)	**D**	380 VY/220VD 60 Hz	**R**
500 VD 50 Hz (mid range value)	**E**	400 VY 50 Hz (mid range value)	**S**
500 VY 50 Hz (mid range value)	**F**	660 VD 50 Hz	**T**
415 VY 50 Hz	**G**	690 VD 50 Hz (mid range value)	**U**
415 VD 50 Hz	**H**	220 VDD/440 VD Hz (Manilla)	**V**
690 VY 60 Hz	**J**	660 VY 60 Hz	**W**
440 VD 50 Hz (mid range value)	**K**	other	**X**
230 VYY/460 VY 60 Hz	**M**	600 VD 60 Hz	**Y**
460 VY/230 VD 60 Hz	**N**	575 VD 60 Hz	**Z**

Motors for other voltages

Motors wound for a given voltage at 50 Hz can also be used for other voltages. Recalculation factors for current and torque values are given below; efficiency, power factor and speed remain approximately the same.
Guaranteed values available on request.

Motor wound for	230V		400V		500V		690 V	
Connected to 50 Hz	220V	230V	380V	415V	500V	550V	660V	690V
% of values at 400V, 50 Hz								
Output	100	100	100	100	100	100	100	100
I_N	182	174	105	98	80	75	61	58
I_S/I_N	90	100	90	106	100	119	90	100
T_S/T_N	90	100	90	106	100	119	90	100
T_{max}/T_N	90	100	90	106	100	119	90	100

Note! The table above not valid for M2AA 160-250.

ABB reserves the right to change the design, technical specification and dimensions without prior notice.

General purpose aluminum motors

Technical data for totally enclosed squirrel cage three phase motors

IP 55, IC 411; Insulation class F, temperature rise class B

Output kW		Type designation	Product code	Speed r/min	Efficiency Full load 100%	Efficiency 3/4 load 75%	Power factor cos φ 100 %	Current I_N A	Current I_s/I_N	Torque T_N Nm	Torque T_s/T_N	Torque T_{max}/T_N	Moment of inertia J=1/4 GD² kgm²	Weight kg	Sound pressure level LP dB(A)
2-poles = 3000 r/min				**400 V 50 Hz**									**Basic design**	EFF 2	
0.09		**M2VA 56 A**	3GVA 051 001-••A	2820	59.8	53.3	0.69	0.32	3.9	0.31	2.9	2.7	0.00011	3.2	48
0.12		**M2VA 56 B**	3GVA 051 002-••A	2840	67.2	63.8	0.64	0.41	4.1	0.41	3.2	2.8	0.00012	3.4	48
0.18		**M2VA 63 A**	3GVA 061 001-••C	2820	73.7	70.6	0.64	0.56	4.2	0.62	3.5	3.1	0.00013	3.9	54
0.25		**M2VA 63 B**	3GVA 061 002-••C	2810	77.5	75.8	0.71	0.66	4.5	0.87	3.6	3.3	0.00016	4.4	54
0.37		**M2VA 71 A**	3GVA 071 001-••C	2840	77.1	76.5	0.72	1.00	5.5	1.25	3.8	3.9	0.0004	5.5	58
0.55		**M2VA 71 B**	3GVA 071 002-••C	2830	79.2	78.2	0.76	1.35	5.7	1.86	3.6	3.7	0.00045	6.5	58
0.75		**M2VA 80 A**	3GVA 081 001-••B	2870	81.2	79.3	0.75	1.80	6.2	2.49	2.9	3.6	0.000722	9	60
1.1		**M2VA 80 B**	3GVA 081 002-••B	2850	81.4	79.5	0.78	2.50	6.1	3.69	2.3	3.5	0.000763	11	60
1.5		**M2AA 90 S**	3GAA 091 001-••E	2870	80.1	76.2	0.82	3.35	5.5	5	2.4	3.0	0.0019	13	63
2.2		**M2AA 90 L**	3GAA 091 002-••E	2885	83.6	83.9	0.87	4.37	6.0	7.5	2.5	3.0	0.0024	16	63
3		**M2AA 100 L**	3GAA 101 001-••E	2900	86.0	84.1	0.88	5.95	7.5	10	2.7	3.6	0.0041	21	65
4		**M2AA 112 M**	3GAA 111 001-••A	2850	86.0	86.0	0.91	7.40	7.5	13.4	2.8	3.0	0.01	25	63
5.5		**M2AA 132 SA**	3GAA 131 001-••A	2855	86.0	86.0	0.88	10.5	6.8	18.3	2.7	3.6	0.014	37	69
7.5		**M2AA 132 SB**	3GAA 131 002-••A	2855	87.0	87.0	0.90	13.9	7.2	25	3.2	3.8	0.016	42	69
11		**M2AA 160 MA**	3GAA 161 111-••A	2915	88.4	88.9	0.89	20.5	6.1	36	2.1	2.5	0.039	73	73
15		**M2AA 160 M**	3GAA 161 112-••A	2900	89.5	89.9	0.90	27	6.1	49.4	2.4	2.6	0.047	84	73
18.5		**M2AA 160 L**	3GAA 161 113-••A	2915	90.2	90.5	0.91	33	6.8	61	2.6	3.0	0.053	94	73
22		**M2AA 180 M**	3GAA 181 111-••A	2925	91.2	91.3	0.89	39	7.9	72	2.8	3.2	0.06	111	75
30		**M2AA 200 LA**	3GAA 201 011-••A	2945	92.0	92.0	0.88	53	7.9	97	3.0	3.7	0.094	139	75
37		**M2AA 200 L**	3GAA 201 012-••A	2945	92.8	92.9	0.89	65	8.2	120	3.1	3.6	0.115	170	75
45		**M2AA 225 M**	3GAA 221 011-••A	2940	93.0	93.0	0.88	80	7.7	146	2.8	3.0	0.21	209	75
55		**M2AA 250 M**	3GAA 251 011-••A	2960	93.5	93.8	0.90	95	7.3	177	2.8	3.0	0.31	277	74
2-poles = 3000 r/min				**400 V 50 Hz**									**Basic design**	EFF 1	
1.1		**M3VA 80 C**	3GVA 081 313-••B	2850	82.8	82.6	0.85	2.25	8.1	3.69	4.2	3.5	0.001093	11	60
1.5		**M3AA 90 L**	3GAA 091 312-••E	2900	85.9	86.5	0.87	3.20	7.7	5	2.7	3.6	0.0024	16	63
2.2		**M3AA 90 LB**	3GAA 091 313-••E	2880	85.8	87.1	0.87	4.40	7.4	7.3	3.0	3.6	0.0027	18	63
3		**M3AA 100 LB**	3GAA 101 312-••E	2920	87.6	87.5	0.86	5.90	10.0	9.9	3.9	4.9	0.005	25	62
4		**M3AA 112 M**	3GAA 111 022-••C	2860	87.7	89.4	0.93	7.1	7.5	13.4	2.6	3.4	0.012	33	63
5.5		**M3AA 132 SA**	3GAA 131 023-••C	2900	88.6	89.6	0.88	10.3	9.7	18.1	3.8	4.3	0.016	42	69
7.5		**M3AA 132 SB**	3GAA 131 024-••C	2915	90.9	91.3	0.9	13.3	11.0	24.6	5.1	5.2	0.022	56	69
11		**M3AA 160 MA**	3GAA 161 101-••C	2930	91.0	91.2	0.88	20	6.2	36	2.1	2.8	0.039	73	69
15		**M3AA 160 M**	3GAA 161 102-••C	2920	91.3	91.7	0.9	27	6.4	49	2.3	2.7	0.047	84	69
18.5		**M3AA 160 L**	3GAA 161 103-••C	2920	92.4	93.1	0.91	32	7.2	61	2.6	2.9	0.053	94	69
22		**M3AA 180 M**	3GAA 181 101-••C	2930	92.8	93.3	0.89	39	7.2	71	2.7	3.0	0.077	119	69
30		**M3AA 200 MLA**	3GAA 201 001-••C	2955	93.2	93.2	0.88	53	8.5	97	2.9	3.1	0.15	175	72
37		**M3AA 200 MLB**	3GAA 201 002-••C	2950	93.6	93.7	0.89	64	7.2	120	2.3	2.9	0.18	200	72
45		**M3AA 225 SMB**	3GAA 221 001-••C	2960	94.1	93.9	0.88	79	7.7	145	2.5	2.9	0.26	235	74
55		**M3AA 250 SMA**	3GAA 251 001-••C	2970	94.2	93.8	0.89	95	7.9	177	2.4	3.0	0.49	285	75
75		**M3AA 280 SMA**	3GAA 281 001-••C	2970	95.1	95.2	0.9	127	8.2	241	2.4	3.1	0.57	330	75
2-poles = 3000 r/min				**400 V 50 Hz**									**High-output design**		
0.37		**M2VA 63 BB**	3GVA 061 003-••C	2800	73.6	73.1	0.81	0.9	3.5	1.29	2.3	2.2	0.00036	4.9	54
0.68		**M2VA 71 BB**	3GVA 071 003-••C	2800	78.9	77.4	0.82	1.59	5.2	2.33	3.2	3.3	0.00045	6.5	58
0.75		**M2VA 71 BC**	3GVA 071 004-••C	2800	78.5	77.9	0.85	1.7	5.1	2.57	3.1	3.2	0.00045	6.5	58
1.5		**M2VA 80 C**	3GVA 081 003-••B	2840	82.4	82.2	0.83	3.16	5.5	5.13	2.8	3.1	0.001093	11.5	60
2.7	1)	**M3AA 90 LB**	3GAA 091 003-••E	2860	80.7	83.5	0.86	5.7	7.0	9	2.6	3.0	0.0027	18	68
4	1)	**M3AA 100 LB**	3GAA 101 002-••E	2900	85.0	84.3	0.86	8.1	7.5	13	2.7	3.6	0.005	25	68
5.5	1)	**M3AA 112 MB**	3GAA 111 002-••C	2855	86.6	87.9	0.93	9.9	7.3	18.4	2.6	3.4	0.012	33	63
9.2	1)	**M3AA 132 SBB**	3GAA 131 004-••C	2840	86.8	88.3	0.92	16.8	8.5	31	3.3	3.6	0.02	50	69
11	1)	**M3AA 132 SC**	3GAA 131 003-••C	2835	87.9	89.2	0.93	19.6	7.7	37	3.0	3.2	0.022	56	69
22	1)	**M3AA 160 LB**	3GAA 161 104-••C	2920	92.0	93.0	0.91	38	6.9	72	2.3	2.9	0.058	100	69
30		**M3AA 180 LB**	3GAA 181 102-••C	2945	93.7	94.0	0.89	53	7.8	97	2.7	3.0	0.092	137	70
45		**M3AA 200 MLC**	3GAA 201 003-••C	2950	94.1	94.5	0.89	78	8.2	146	3.0	3.2	0.19	205	72
55	1)	**M3AA 200 MLD**	3GAA 201 004-••C	2940	94.0	94.4	0.89	95	7.9	179	3.1	3.1	0.2	215	2)
55		**M3AA 225 SMC**	3GAA 221 002-••C	2960	94.5	94.6	0.89	95	7.3	177	2.8	3.0	0.29	260	74
80	1)	**M3AA 225 SMD**	3GAA 221 003-••C	2960	94.7	94.7	0.86	143	7.5	258	2.9	3.1	0.3	275	74
75		**M3AA 250 SMB**	3GAA 251 002-••C	2970	95.0	94.9	0.90	127	8.6	241	2.7	3.3	0.57	375	75
95	1)	**M3AA 250 SMC**	3GAA 251 003-••C	2965	95.4	95.6	0.90	160	8.0	306	2.6	3.1	0.59	345	75
90	1)	**M3AA 280 SMB**	3GAA 281 002-••C	2970	95.4	94.8	0.90	152	8.3	290	2.7	3.4	0.59	390	75

[1] Temperature rise class F

Efficiency classes fixed for ranges 1.1 to 90 kW (available only by 2- and 4-poles).

General purpose aluminum motors

Technical data for totally enclosed squirrel cage three phase motors

IP 55, IC 411; Insulation class F, temperature rise class B

Output kW	Type designation	Product code	Speed r/min	Efficiency Full load 100%	Efficiency 3/4 load 75%	Power factor cos φ 100 %	Current I_N A	Current I_s/I_N	Torque T_N Nm	Torque T_s/T_N	Torque T_{max}/T_N	Moment of inertia J=1/4 GD² kgm²	Weight kg	Sound pressure level LP dB(A)
4-poles = 1500 r/min			**400 V 50 Hz**							**Basic design**			EFF 2	
0.06	**M2VA 56 A**	3GVA 052 001-••A	1340	51.1	45.8	0.67	0.26	2.5	0.43	2.2	2.2	0.00017	3.2	36
0.09	**M2VA 56 B**	3GVA 052 002-••A	1370	55.5	50.2	0.62	0.38	2.8	0.63	2.9	2.9	0.00018	3.4	36
0.12	**M2VA 63 A**	3GVA 062 001-••C	1400	63.7	58.4	0.59	0.46	3.1	0.82	2.6	2.6	0.00019	4	40
0.18	**M2VA 63 B**	3GVA 062 002-••C	1380	65.6	62.1	0.64	0.63	3.1	1.25	2.5	2.6	0.00026	4.5	40
0.25	**M2VA 71 A**	3GVA 072 001-••C	1410	70.4	69.1	0.71	0.74	4.3	1.71	2.7	2.9	0.00066	5.5	45
0.37	**M2VA 71 B**	3GVA 072 002-••C	1420	74.6	72.1	0.69	1.05	4.4	2.51	2.6	2.8	0.00089	6.5	45
0.55	**M2VA 80 A**	3GVA 082 001-••B	1390	75.3	73.1	0.76	1.40	4.6	3.75	2.6	2.9	0.001257	9	50
0.75	**M2VA 80 B**	3GVA 082 002-••B	1410	78.2	75.6	0.74	1.90	4.7	5.08	3.5	3.9	0.001565	10.5	50
1.1	**M2AA 90 S**	3GAA 092 001-••E	1410	77.5	76.4	0.81	2.59	5.0	7.5	2.2	2.7	0.0032	13	50
1.5	**M2AA 90 L**	3GAA 092 002-••E	1420	80.3	78.1	0.79	3.45	5.0	10	2.4	2.9	0.0043	16	50
2.2	**M2AA 100 LA**	3GAA 102 001-••E	1430	83.0	82.7	0.81	4.80	5.5	15	2.4	2.9	0.0069	21	64
3	**M2AA 100 LB**	3GAA 102 002-••E	1430	85.0	83.9	0.81	6.48	5.5	20	2.5	2.9	0.0082	24	66
4	**M2AA 112 M**	3GAA 112 001-••A	1435	84.5	85.5	0.8	8.60	7.0	27	2.8	3.0	0.015	27	56
5.5	**M2AA 132 S**	3GAA 132 001-••A	1450	87.0	87.0	0.83	11.10	6.9	36	2.2	3.0	0.031	40	59
7.5	**M2AA 132 M**	3GAA 132 002-••A	1450	88.0	88.0	0.83	14.80	7.9	49	2.5	3.2	0.038	48	59
11	**M2AA 160 M**	3GAA 162 111-••A	1460	89.1	89.8	0.81	22	6.5	72	2.7	2.7	0.067	75	62
15	**M2AA 160 L**	3GAA 162 112-••A	1460	90.4	91.0	0.82	29	7.1	98	2.7	3.3	0.088	92	62
18.5	**M2AA 180 M**	3GAA 182 111-••A	1460	91.1	91.5	0.81	37	7.6	121	3.1	3.5	0.102	110	64
22	**M2AA 180 L**	3GAA 182 112-••A	1460	91.8	92.3	0.82	42	7.9	144	3.1	3.8	0.127	128	64
30	**M2AA 200 L**	3GAA 202 011-••A	1470	92.0	92.1	0.8	59	7.8	195	3.0	3.4	0.225	177	67
37	**M2AA 225 S**	3GAA 222 011-••A	1475	92.8	93.0	0.85	68	6.8	240	3.0	3.1	0.35	216	68
45	**M2AA 225 M**	3GAA 222 012-••A	1475	93.0	93.1	0.84	84	8.1	291	3.5	3.2	0.41	237	68
55	**M2AA 250 M**	3GAA 252 011-••A	1475	93.7	94.3	0.84	98	6.8	356	2.5	2.6	0.5	286	66
4-poles = 1500 r/min			**400 V 50 Hz**							**Basic design**			EFF 1	
1.1	**M3AA 90 L**	3GAA 092 312-••E	1420	83.9	84.3	0.8	2.40	6.1	7.4	2.9	3.4	0.0043	16	50
1.5	**M3AA 100 LA**	3GAA 102 311-••E	1440	85.6	85.5	0.82	3.20	6.9	10	2.8	3.4	0.0069	21	54
2.2	**M3AA 100 LC**	3GAA 102 313-••E	1450	86.8	86.5	0.77	4.80	8.5	14.5	4.0	4.6	0.009	25	54
3	**M3AA 112 MA**	3GAA 112 021-••C	1455	87.5	87.8	0.81	6.20	7.9	19.7	2.7	3.7	0.018	34	56
4	**M3AA 112 M**	3GAA 112 022-••C	1455	88.3	88.6	0.76	8.60	8.5	26.3	3.3	4.3	0.018	34	56
5.5	**M3AA 132 S**	3GAA 132 023-••C	1460	89.3	90.5	0.84	10.60	7.5	36	2.6	3.1	0.038	48	59
7.5	**M3AA 132 M**	3GAA 132 024-••C	1450	90.1	91.4	0.87	14.00	7.8	49	2.2	3.1	0.048	59	59
11	**M3AA 160 M**	3GAA 162 101-••C	1460	92.0	92.7	0.81	21.50	7.6	72	2.9	3.4	0.091	94	62
15	**M3AA 160 L**	3GAA 162 102-••C	1460	91.8	92.5	0.82	29.00	8.1	98	3.3	3.6	0.102	103	62
18.5	**M3AA 180 M**	3GAA 182 101-••C	1470	92.3	92.9	0.84	35	7.0	120	2.9	2.9	0.161	124	62
22	**M3AA 180 L**	3GAA 182 102-••C	1470	93.1	93.9	0.85	40	7.0	143	2.6	2.8	0.225	161	63
30	**M3AA 200 MLB**	3GAA 202 001-••C	1475	93.4	94.0	0.84	55	7.5	194	2.5	2.8	0.34	205	63
37	**M3AA 225 SMA**	3GAA 222 001-••C	1480	93.6	93.8	0.84	68	7.4	239	2.9	3.0	0.37	215	66
45	**M3AA 225 SMB**	3GAA 222 002-••C	1480	94.2	94.4	0.83	83	7.6	291	2.8	3.0	0.42	230	66
55	**M3AA 250 SMA**	3GAA 252 001-••C	1480	94.6	94.9	0.86	98	7.6	355	3.1	3.0	0.72	275	67
75	**M3AA 280 SMA**	3GAA 282 001-••C	1480	94.8	95.3	0.86	132	7.1	486	2.7	3.0	0.88	380	67
90	**M3AA 280 SMB**	3GAA 282 002-••C	1475	95.0	95.3	0.87	157	7.7	583	3.3	3.2	0.95	405	67
4-poles = 1500 r/min			**400 V 50 Hz**							**High-output design**				
0.25	**M2VA 63 BB**	3GVA 062 003-••C	1370	70.3	67.4	0.67	0.78	3.2	1.75	2.5	2.1	0.0003	5	40
0.45	**M2VA 71 BB**	3GVA 072 003-••C	1390	75.5	75.3	0.76	1.15	4.1	3.11	2.1	2.3	0.00089	6.5	45
0.55	**M2VA 71 C**	3GVA 072 004-••C	1410	77.3	76.9	0.73	1.45	4.8	3.74	2.7	2.9	0.0011	7	45
0.95	**M2VA 80 C**	3GVA 082 003-••B	1410	78.9	77.9	0.75	2.35	4.3	6.44	2.9	3.3	0.001948	11	50
1.1	**M2VA 80 C**	3GVA 082 004-••B	1390	74.7	76.6	0.77	2.80	4.3	7.8	3.1	2.3	0.001948	11	50
1.85 [1]	**M3AA 90 L**	3GAA 092 003-••E	1390	79.5	78.1	0.8	4.40	4.5	13	2.2	2.4	0.0043	16	50
2.2 [1]	**M3AA 90 LB**	3GAA 092 004-••E	1390	80.3	81.0	0.83	4.85	4.5	15	2.2	2.4	0.0048	17	50
4 [1]	**M3AA 100 LC**	3GAA 102 003-••E	1420	81.0	81.7	0.82	8.65	5.5	27	2.5	2.8	0.009	25	60
5.5 [1]	**M3AA 112 MB**	3GAA 112 002-••C	1425	85.3	86.5	0.83	11.40	6.7	37	2.5	3.2	0.018	34	56
9.2 [1]	**M3AA 132 MBA**	3GAA 132 004-••C	1445	87.8	89.2	0.87	17.50	7.2	61	2.7	2.7	0.048	59	59
11 [1]	**M3AA 132 MB**	3GAA 132 003-••C	1450	88.6	89.9	0.86	21	7.4	73	2.5	2.7	0.048	59	59
18.5 [1]	**M3AA 160 LB**	3GAA 162 103-••C	1450	90.5	92.0	0.84	36	6.6	122	2.6	3.0	0.102	103	63
30 [1]	**M3AA 180 LB**	3GAA 182 103-••C	1465	92.5	93.3	0.84	56	6.8	196	2.5	2.8	0.225	161	63
37	**M3AA 200 MLB**	3GAA 202 002-••C	1475	93.8	94.2	0.84	68	7.4	240	2.9	2.7	0.34	205	63
48 [1]	**M3AA 200 MLC**	3GAA 202 003-••C	1470	93.6	94.1	0.84	89	8.1	311	4.4	3.2	0.38	270	63
55	**M3AA 225 SMC**	3GAA 222 003-••C	1480	94.6	95.0	0.84	100	7.5	356	3.5	3.0	0.49	265	66
73 [1]	**M3AA 225 SMD**	3GAA 222 004-••C	1475	94.2	94.5	0.85	132	8.1	473	3.9	3.2	0.56	290	66
75	**M3AA 250 SMB**	3GAA 252 002-••C	1480	94.6	95.1	0.86	132	6.9	486	2.6	3.0	0.88	335	67
95 [1]	**M3AA 250 SMC**	3GAA 252 003-••C	1475	94.5	95.3	0.88	165	7.3	616	2.6	3.1	0.95	360	67

[1] Temperature rise class F

Efficiency classes fixed for ranges 1.1 to 90 kW (available only by 2- and 4-poles).

General purpose aluminum motors

Technical data for totally enclosed squirrel cage three phase motors

IP 55, IC 411; Insulation class F, temperature rise class B

Output kW	Type designation	Product code	Speed r/min	Efficiency Full load 100%	Efficiency 3/4 load 75%	Power factor cos φ 100 %	Current I_N A	Current I_s/I_N	Torque T_N Nm	T_s/T_N	T_{max}/T_N	Moment of inertia J=1/4 GD² kgm²	Weight kg	Sound pressure level LP dB(A)
6-poles = 1000 r/min			**400 V 50 Hz**											**Basic design**
0.09	**M2VA 63 A**	3GVA 063 001-••C	910	47.1	42.5	0.56	0.51	2.1	0.95	2.1	2.1	0.0002	4	38
0.12	**M2VA 63 B**	3GVA 063 002-••C	910	57.5	54.0	0.58	0.54	2.1	1.27	2.1	2.1	0.00027	4.5	38
0.18	**M2VA 71 A**	3GVA 073 001-••C	920	61.1	57.7	0.69	0.64	2.9	1.88	2.1	2.2	0.00063	5.5	42
0.25	**M2VA 71 B**	3GVA 073 002-••C	920	64.9	62.3	0.65	0.86	3.2	2.61	2.5	2.7	0.00081	6.5	42
0.37	**M2VA 80 A**	3GVA 083 001-••B	925	72.9	70.8	0.72	1.04	3.8	3.82	3.1	3.4	0.001842	9	47
0.55	**M2VA 80 B**	3GVA 083 002-••B	925	73.3	71.9	0.71	1.55	3.4	5.68	2.9	3.1	0.002176	10	47
0.75	**M3AA 90 S**	3GAA 093 001-••E	930	71.5	70.7	0.67	2.36	4.0	7.5	1.9	2.3	0.0032	13	44
1.1	**M3AA 90 L**	3GAA 093 002-••E	930	74.4	72.5	0.69	3.25	4.0	11	2.1	2.4	0.0043	16	44
1.5	**M3AA 100 L**	3GAA 103 001-••E	950	80.0	77.0	0.71	3.92	4.5	15	1.9	2.3	0.0082	23	49
2.2	**M3AA 112 M**	3GAA 113 001-••C	940	80.5	81.0	0.74	5.40	5.6	22	2.1	2.7	0.015	27	54
3	**M3AA 132 S**	3GAA 133 001-••C	960	84.5	84.8	0.75	6.90	6.5	30	2.1	3.0	0.031	39	61
4	**M3AA 132 MA**	3GAA 133 002-••C	960	85.5	86.1	0.78	8.70	7.1	40	2.6	2.8	0.038	46	61
5.5	**M3AA 132 MB**	3GAA 133 003-••C	955	86.0	87.0	0.78	11.90	6.6	55	2.1	2.8	0.045	54	61
7.5	**M3AA 160 M**	3GAA 163 101-••C	970	89.3	90.4	0.79	15.40	6.6	74	1.9	2.6	0.089	88	59
11	**M3AA 160 L**	3GAA 163 102-••C	970	89.8	90.5	0.78	23	6.9	109	2.1	3.4	0.107	102	59
15	**M3AA 180 L**	3GAA 183 101-••C	970	90.8	91.5	0.78	31	6.8	147	2.0	3.3	0.217	151	59
18.5	**M3AA 200 MLA**	3GAA 203 001-••C	985	91.1	91.7	0.81	36	7.0	180	2.7	2.5	0.37	165	63
22	**M3AA 200 MLB**	3GAA 203 002-••C	980	91.7	92.2	0.81	43	6.8	214	2.9	3.0	0.43	185	63
30	**M3AA 225 SMB**	3GAA 223 001-••C	985	92.8	93.0	0.83	56	7.2	291	3.1	2.9	0.64	225	63
37	**M3AA 250 SMA**	3GAA 253 001-••C	985	93.4	93.7	0.83	69	7.3	358	3.1	2.8	1.16	280	63
45 [1]	**M3AA 280 SMA**	3GAA 283 001-••C	985	93.4	93.7	0.84	83	7.2	436	3.2	2.8	1.49	375	63
6-poles = 1000 r/min			**400 V 50 Hz**											**High-output design**
0.15	**M2VA 63 BB**	3GVA 063 003-••C	900	56.9	52.1	0.54	0.74	2.2	1.61	2.2	2.3	0.00032	5	38
0.32	**M2VA 71 C**	3GVA 073 003-••C	920	64.8	61.6	0.63	1.15	3.2	3.33	2.6	2.8	0.0011	7	42
0.37	**M2VA 71 C**	3GVA 073 004-••C	900	60.1	60.4	0.7	1.20	2.6	4.1	2.2	2.0	0.0011	7	42
0.75	**M2VA 80 C**	3GVA 083 003-••B	920	67.9	70.5	0.76	2.10	3.4	8.1	2.4	2.2	0.002576	10	47
1.3 [1]	**M3AA 90 LB**	3GAA 093 003-••E	910	69.0	69.0	0.71	3.85	4.0	13.5	1.9	2.2	0.0048	18	44
2.2 [1]	**M3AA 100 LC**	3GAA 103 002-••E	940	77.0	72.8	0.71	5.90	4.5	22	1.9	2.3	0.009	26	49
3 [1]	**M3AA 112 MB**	3GAA 113 002-••C	935	80.0	81.2	0.76	7.20	5.5	31	2.5	2.7	0.018	33	54
6.3 [1]	**M3AA 132 MC**	3GAA 133 004-••C	960	84.9	85.0	0.75	14.50	7.3	63	2.3	3.1	0.049	59	61
14 [1]	**M3AA 160 LB**	3GAA 163 103-••C	960	89.8	90.1	0.77	29.50	7.0	138	2.5	3.1	0.127	117	62
18.5 [1]	**M3AA 180 LB**	3GAA 183 102-••C	965	90.7	91.7	0.8	37	6.1	183	2.1	2.5	0.237	160	59
30 [1]	**M3AA 200 MLC**	3GAA 203 003-••C	980	91.9	92.5	0.81	56	7.1	293	3.3	2.9	0.49	200	63
37	**M3AA 225 SMC**	3GAA 223 002-••C	985	92.8	93.4	0.83	69	6.9	360	3.0	2.9	0.75	252	63
45 [1]	**M3AA 250 SMB**	3GAA 253 002-••C	985	93.4	93.7	0.84	83	7.2	436	3.2	2.8	1.49	320	63

[1] Temperature rise class F

General purpose aluminum motors

Technical data for totally enclosed squirrel cage three phase motors

IP 55, IC 411; Insulation class F, temperature rise class B

Output kW	Type designation	Product code	Speed r/min	Efficiency Full load 100%	Efficiency 3/4 load 75%	Power factor cos φ 100 %	Current I_N A	Current I_s/I_N	Torque T_N Nm	T_s/T_N	T_{max}/T_N	Moment of inertia J=1/4 GD² kgm²	Weight kg	Sound pressure level LP dB(A)
8-poles = 750 r/min			**400 V 50 Hz**									**Basic design**		
0.055	**M2VA 63 B**	3GVA 064 002-••C	680	38.3	31.8	0.48	0.45	1.8	0.78	2.1	2.1	0.00027	4.5	36
0.09	**M2VA 71 A**	3GVA 074 001-••C	690	45.8	37.5	0.57	0.52	2.2	1.25	2.3	2.3	0.00063	5.5	40
0.12	**M2VA 71 B**	3GVA 074 002-••C	690	46.4	38.1	0.55	0.69	2.2	1.67	2.5	2.5	0.00081	6.5	40
0.18	**M2VA 80 A**	3GVA 084 001-••B	700	59.9	54.5	0.60	0.75	3.1	2.46	3.2	3.6	0.001842	9	45
0.25	**M2VA 80 B**	3GVA 084 002-••B	700	70.7	67.4	0.62	0.85	3.1	3.52	2.9	3.1	0.002176	10	45
0.37	**M3AA 90 S**	3GAA 094 001-••E	700	61.5	43.4	0.56	1.60	3.0	5	1.9	2.4	0.0032	13	43
0.55	**M3AA 90 L**	3GAA 094 002-••E	690	62.9	56.4	0.57	2.35	3.0	7.5	1.7	2.1	0.0043	16	43
0.75	**M3AA 100 LA**	3GAA 104 001-••E	700	72.0	63.6	0.59	2.55	3.5	10	2.1	2.7	0.0069	20	46
1.1	**M3AA 100 LB**	3GAA 104 002-••E	700	73.0	68.8	0.64	3.35	3.5	15	2.1	2.7	0.0082	23	46
1.5	**M3AA 112 M**	3GAA 114 001-••C	695	74.5	74.6	0.65	4.50	4.1	21	1.9	2.5	0.016	28	52
2.2	**M3AA 132 S**	3GAA 134 001-••C	720	80.5	80.2	0.67	5.90	5.3	29	1.9	2.5	0.038	46	56
3	**M3AA 132 M**	3GAA 134 002-••C	720	82.0	82.0	0.68	7.80	5.5	40	2.4	2.6	0.045	53	56
4	**M3AA 160 MA**	3GAA 164 101-••C	715	84.1	84.7	0.69	10.00	5.1	53	2.1	2.6	0.072	75	59
5.5	**M3AA 160 M**	3GAA 164 102-••C	710	84.7	85.6	0.70	13.40	5.5	74	2.4	2.6	0.091	88	59
7.5	**M3AA 160 L**	3GAA 164 103-••C	715	86.3	87.3	0.70	18.10	5.4	100	2.4	2.7	0.131	118	59
11	**M3AA 180 L**	3GAA 184 101-••C	720	89.6	90.3	0.76	23.50	5.7	146	2.1	2.5	0.224	147	59
15	**M3AA 200 MLA**	3GAA 204 001-••C	740	91.1	91.6	0.82	29	7.5	196	3.0	3.2	0.45	175	60
18.5	**M3AA 225 SMA**	3GAA 224 001-••C	730	91.1	91.6	0.79	37	6.8	242	2.8	3.1	0.61	210	63
22	**M3AA 225 SMB**	3GAA 224 002-••C	730	91.5	92.2	0.77	45	6.4	287	2.4	2.6	0.68	225	63
30	**M3AA 250 SMA**	3GAA 254 001-••C	735	92.8	93.1	0.79	59	7.3	389	2.2	2.6	1.25	280	63
37	**M3AA 280 SMA**	3GAA 284 001-••C	735	93.0	93.3	0.81	74	7.4	479	2.0	2.6	1.52	375	63
8-poles = 750 r/min			**400 V 50 Hz**									**High-output design**		
0.18	**M2VA 71 C**	3GVA 074 003-••C	680	51.3	49.9	0.61	0.80	2.2	2.6	2.5	2.2	0.0011	7	40
0.37	**M2VA 80 C**	3GVA 084 003-••B	690	64.6	65.3	0.69	1.20	3.0	5.3	2.3	2.1	0.002576	11	45
0.75 [1)]	**M3AA 90 LB**	3GAA 094 003-••E	680	64.0	60.0	0.60	3.10	3.0	10	1.8	2.0	0.0048	18	43
1.5 [1)]	**M3AA 100 LC**	3GAA 104 003-••E	670	71.0	65.9	0.70	4.40	3.3	21	1.8	2.2	0.009	26	46
1.9 [1)]	**M3AA 112 MB**	3GAA 114 002-••C	690	74.0	74.8	0.67	5.60	4.3	26.5	2.0	2.6	0.018	33	52
3.8 [1)]	**M3AA 132 MB**	3GAA 134 003-••C	710	80.5	80.7	0.69	9.90	5.2	51	2.3	2.6	0.049	59	56
8.5 [1)]	**M3AA 160 LB**	3GAA 164 104-••C	700	85.1	85.7	0.70	21.00	5.3	114	2.3	2.6	0.131	118	62
15 [1)]	**M3AA 180 LB**	3GAA 184 102-••C	720	88.7	89.6	0.76	32.50	6.0	199	2.4	2.6	0.24	155	62
18.5	**M3AA 200 MLB**	3GAA 204 002-••C	735	91.4	91.8	0.81	36	7.3	241	2.6	3.1	0.54	200	60
30 [1)]	**M3AA 225 SMC**	3GAA 224 003-••C	735	90.5	91.3	0.79	64	6.7	391	2.5	3.0	0.8	255	63
37	**M3AA 250 SMB**	3GAA 254 002-••C	735	93.0	93.3	0.81	74	7.4	479	2.0	2.6	1.52	320	63

[1)] Temperature rise class F

General purpose steel motors

Technical data for totally enclosed squirrel cage three phase motors

IP 55, IC 411; Insulation class F, temperature rise class B

Output kW	Type designation	Product code	Speed r/min	Efficiency Full load 100%	Efficiency 3/4 load 75%	Power factor cos φ 100 %	Current I_N A	Current I_s/I_N	Torque T_N Nm	T_s/T_N	T_{max}/T_N	Moment of inertia J=1/4 GD² kgm²	Weight kg	Sound pressure level LP dB(A)
2-poles = 3000 r/min			**400 V 50 Hz**							**Basic design**			EFF 2	
75	**M2CA 280 SA**	3GCA 281 110-••A	2977	94.9	94.6	0.88	131	7.5	241	2.3	3.3	0.8	480	77
90	**M2CA 280 SMA**	3GCA 281 210-••A	2975	95.1	94.9	0.90	152	7.6	289	2.3	2.9	0.9	545	77
110	**M2CA 315 SA**	3GCA 311 110-••A	2982	95.1	94.4	0.86	194	7.6	352	2.0	3.0	1.2	695	80
132	**M2CA 315 SMA**	3GCA 311 210-••A	2982	95.4	94.9	0.88	228	7.4	423	2.2	3.0	1.4	770	80
160	**M2CA 315 MB**	3GCA 311 320-••A	2981	96.1	95.6	0.89	269	7.5	513	2.3	3.0	1.7	840	80
200	**M2CA 315 LA**	3GCA 311 510-••A	2978	96.3	95.9	0.90	334	7.8	641	2.6	3.0	2.1	975	80
200	**M2CA 355 SA**	3GCA 351 110-••C	2978	95.5	95.2	0.89	338	6.4	641	1.3	2.6	2.5	1200	83
250	**M2CA 355 MA**	3GCA 351 310-••C	2983	96.1	95.6	0.89	422	7.2	800	1.4	3.0	2.7	1260	83
280	**M2CA 355 MB**	3GCA 351 320-••C	2981	96.1	95.7	0.89	472	6.8	897	1.3	2.8	2.7	1260	83
315	**M2CA 355 LA**	3GCA 351 510-••C	2980	96.4	96.1	0.89	535	7.0	1009	2.1	3.0	3.4	1480	83
355	**M2CA 355 LB**	3GCA 351 520-••C	2983	96.5	96.2	0.88	603	7.7	1136	2.1	2.9	3.5	1520	83
400	**M2CA 400 MLA**	3GCA 401 410-••C	2985	96.8	96.5	0.88	675	7.2	1280	1.4	2.6	6.3	2050	85
450	**M2CA 400 MLB**	3GCA 401 420-••C	2987	96.9	96.7	0.90	743	7.7	1438	1.7	3.0	6.9	2150	85
500	**M2CA 400 LKA**	3GCA 401 810-••C	2987	97.0	96.8	0.90	825	8.0	1598	2.0	3.2	7.8	2450	85
560	**M2CA 400 LKB**	3GCA 401 820-••C	2988	97.2	97.0	0.89	940	7.8	1790	2.1	3.4	7.9	2500	85
2-poles = 3000 r/min			**400 V 50 Hz**							**High-output design**				
110	**M2CA 280 MB**	3GCA 281 320-••A	2977	95.8	95.5	0.90	184	7.9	353	2.4	3.0	1.15	580	77
132	**M2CA 280 MC**	3GCA 281 330-••A	2976	96.0	95.7	0.91	222	7.7	424	2.6	3.0	1.4	755	77
160	**M2CA 280 MD**	3GCA 281 340-••A	2975	96.2	96.0	0.91	266	7.9	514	2.8	3.1	1.55	810	77
250	**M2CA 315 LB**	3GCA 311 520-••A	2980	96.5	96.4	0.89	422	8.1	801	2.8	2.9	2.65	1230	80
315 [1]	**M2CA 315 LC**	3GCA 311 530-••A	2982	96.7	96.6	0.89	530	8.8	1009	3.2	3.2	3.3	1410	80
4-poles = 1500 r/min			**400 V 50 Hz**							**Basic design**			EFF 2	
75	**M2CA 280 SA**	3GCA 282 110-••A	1483	95.0	94.9	0.84	137	6.8	483	2.4	2.8	1.15	445	68
90	**M2CA 280 SMA**	3GCA 282 210-••A	1484	95.2	95.1	0.85	163	7.1	579	2.7	2.9	1.4	490	68
110	**M2CA 315 SA**	3GCA 312 110-••A	1487	95.4	95.1	0.85	198	6.9	706	2.1	2.8	2	675	71
132	**M2CA 315 SMA**	3GCA 312 210-••A	1486	95.6	95.5	0.85	238	6.7	848	2.2	2.7	2.3	730	71
160	**M2CA 315 MB**	3GCA 312 320-••A	1486	96.0	95.9	0.86	282	7.2	1028	2.4	2.9	2.9	850	71
200	**M2CA 315 LA**	3GCA 312 510-••A	1486	96.2	96.2	0.86	351	7.2	1285	2.5	2.9	3.5	970	71
200	**M2CA 355 SA**	3GCA 352 110-••C	1488	96.0	95.7	0.86	350	7.3	1284	2.0	2.6	4.8	1200	80
250	**M2CA 355 MA**	3GCA 352 310-••C	1489	96.2	96.0	0.86	435	7.5	1603	2.2	2.6	5.7	1320	80
315	**M2CA 355 LA**	3GCA 352 510-••C	1488	96.4	96.2	0.86	550	7.3	2022	2.3	2.8	6.9	1550	80
355	**M2CA 355 LB**	3GCA 352 520-••C	1489	96.6	96.5	0.86	615	7.5	2277	2.4	2.7	6.9	1550	80
400	**M2CA 355 LKD**	3GCA 352 840-••C	1490	96.6	96.3	0.87	684	7.7	2564	2.5	2.8	8.4	1900	85
450	**M2CA 400 MLA**	3GCA 402 410-••C	1491	96.6	96.3	0.87	772	7.4	2882	1.9	2.7	12	2300	85
500	**M2CA 400 MLB**	3GCA 402 420-••C	1491	96.8	96.5	0.86	867	7.8	3202	2.2	2.9	13	2400	85
560	**M2CA 400 LKA**	3GCA 402 810-••C	1491	96.9	96.6	0.85	982	7.4	3587	2.4	3.0	15	2700	85
630	**M2CA 400 LKB**	3GCA 402 820-••C	1491	96.9	96.7	0.87	1077	7.5	4034	2.2	3.0	16	2800	85
4-poles = 1500 r/min			**400 V 50 Hz**							**High-output design**				
110	**M2CA 280 MB**	3GCA 282 320-••A	1483	95.3	95.2	0.86	195	7.5	708	2.7	2.8	1.7	550	68
132	**M2CA 280 MC**	3GCA 282 330-••A	1483	95.6	95.5	0.86	235	7.1	850	2.8	2.9	2.3	775	70
160	**M2CA 280 MD**	3GCA 282 340-••A	1483	95.8	95.7	0.86	283	7.1	1030	2.8	3.1	2.5	820	70
250	**M2CA 315 LB**	3GCA 312 520-••A	1487	96.2	96.2	0.86	442	7.4	1605	2.5	2.9	4.4	1200	78
315	**M2CA 315 LC**	3GCA 312 530-••A	1488	96.5	96.4	0.86	548	7.8	2022	2.6	3.2	5.5	1380	78

[1] Temperature rise class F

Efficiency classes fixed for ranges 1.1 to 90 kW
(available only by 2- and 4-poles).

General purpose steel motors

Technical data for totally enclosed squirrel cage three phase motors

IP 55, IC 411; Insulation class F, temperature rise class B

Output kW		Type designation	Product code	Speed r/min	Efficiency Full load 100%	Efficiency 3/4 load 75%	Power factor cos φ 100 %	Current I_N A	Current I_s/I_N	Torque T_N Nm	T_s/T_N	T_{max}/T_N	Moment of inertia J=1/4 GD² kgm²	Weight kg	Sound pressure level LP dB(A)
6-poles = 1000 r/min				**400 V 50 Hz**									**Basic design**		
45		**M2CA 280 SA**	3GCA 283 110-••A	990	94.1	94.0	0.82	85	6.6	434	2.5	2.5	1.65	440	66
55		**M2CA 280 SMA**	3GCA 283 210-••A	989	94.4	94.3	0.83	102	6.6	531	2.5	2.5	2	475	66
75		**M2CA 315 SA**	3GCA 313 110-••A	992	94.9	94.7	0.80	143	7.1	722	2.3	2.7	2.9	630	72
90		**M2CA 315 SMA**	3GCA 313 210-••A	991	95.3	95.2	0.83	165	7.1	867	2.3	2.7	3.8	720	72
110		**M2CA 315 MB**	3GCA 313 320-••A	991	95.3	95.1	0.83	201	7.3	1060	2.5	2.8	4.5	805	75
132		**M2CA 315 LA**	3GCA 313 510-••A	990	95.4	95.3	0.84	241	6.7	1273	2.4	2.7	5.4	910	75
132		**M2CA 355 SA**	3GCA 353 110-••C	992	95.3	95.0	0.84	238	6.8	1271	2.0	2.4	6.8	1150	79
160		**M2CA 355 SB**	3GCA 353 120-••C	992	95.5	95.3	0.83	290	7.2	1540	2.3	2.5	7.6	1220	79
200		**M2CA 355 MA**	3GCA 353 310-••C	992	95.8	95.6	0.83	363	7.5	1925	2.4	2.6	9	1400	79
250		**M2CA 355 MB**	3GCA 353 320-••C	993	96.0	95.7	0.80	470	7.7	2404	2.9	3.0	10.6	1550	79
315		**M2CA 355 LKD**	3GCA 353 840-••C	992	96.2	96.0	0.82	577	7.4	3032	2.6	2.7	13.2	1900	79
355		**M2CA 400 MLA**	3GCA 403 410-••C	993	96.5	96.4	0.84	630	7.3	3414	2.0	2.4	18	2400	80
400		**M2CA 400 MLB**	3GCA 403 420-••C	994	96.5	96.4	0.84	710	7.6	3843	2.2	2.7	18	2400	80
450	[1]	**M2CA 400 LKA**	3GCA 403 810-••C	994	96.7	96.5	0.83	808	7.8	4323	2.3	2.6	21	2700	80
500	[1]	**M2CA 400 LKB**	3GCA 403 820-••C	994	96.7	96.6	0.83	898	7.7	4803	2.4	2.5	21	2700	80
6-poles = 1000 r/min				**400 V 50 Hz**									**High-output design**		
75		**M2CA 280 MB**	3GCA 283 320-••A	990	94.5	94.4	0.83	139	7.3	723	2.8	2.7	2.6	545	67
90		**M2CA 280 MC**	3GCA 283 330-••A	989	94.9	94.8	0.83	168	7.4	869	2.9	2.9	3.1	815	67
110		**M2CA 280 MD**	3GCA 283 340-••A	990	95.2	95.1	0.83	202	7.9	1061	3.1	3.0	4.1	835	67
8-poles = 750 r/min				**400 V 50 Hz**									**Basic design**		
37		**M2CA 280 SA**	3GCA 284 110-••A	741	93.4	93.1	0.78	74	7.3	477	1.8	3.1	1.85	460	65
45		**M2CA 280 SMA**	3GCA 284 210-••A	741	94.0	93.8	0.78	90	7.6	580	1.9	3.2	2.2	500	65
55		**M2CA 315 SA**	3GCA 314 110-••A	741	94.0	93.7	0.80	107	7.1	710	1.8	2.8	2.9	630	70
75		**M2CA 315 SMA**	3GCA 314 210-••A	740	94.5	94.2	0.81	142	7.1	968	1.8	2.8	3.8	715	70
90		**M2CA 315 MB**	3GCA 314 320-••A	740	94.7	94.5	0.82	169	7.3	1161	1.9	2.8	4.5	800	77
110		**M2CA 315 LA**	3GCA 314 510-••A	740	94.8	94.7	0.83	202	7.0	1420	1.9	2.7	5.4	900	77
110		**M2CA 355 SA**	3GCA 354 110-••C	743	94.9	94.8	0.80	208	6.0	1414	1.0	2.4	6.8	1150	75
132		**M2CA 355 MA**	3GCA 354 310-••C	743	95.1	95.0	0.80	250	6.2	1697	1.0	2.4	7.6	1220	75
160		**M2CA 355 MB**	3GCA 354 320-••C	744	95.3	95.2	0.79	306	6.8	2054	1.2	2.7	9	1400	75
8-poles = 750 r/min				**400 V 50 Hz**									**High-output design**		
55		**M2CA 280 MB**	3GCA 284 320-••A	741	94.4	94.2	0.79	108	7.8	709	1.9	3.2	2.85	575	65

[1] Temperature rise class F

General purpose cast iron motors

Technical data for totally enclosed squirrel cage three phase motors

IP 55, IC 411; Insulation class F, temperature rise class B

Output kW	Type designation	Product code	Speed r/min	Efficiency Full load 100%	Efficiency 3/4 load 75%	Power factor cos φ 100 %	Current I_N A	Current I_s/I_N	Torque T_N Nm	Torque T_s/T_N	Torque T_{max}/T_N	Moment of inertia J=1/4 GD² kgm²	Weight kg	Sound pressure level LP dB(A)
2-poles = 3000 r/min			**400 V 50 Hz**							**Basic design**			EFF 2	
0.37	**M2QA 71 M2A**	3GQA 071 301-••A	2780	70.0	68.0	0.81	0.94	6.1	1.27	2.2	3.0	0.0003	10	56
0.55	**M2QA 71 M2B**	3GQA 071 302-••A	2785	73.0	72.4	0.82	1.33	6.1	1.89	2.2	2.7	0.00037	11	56
0.75	**M2QA 80 M2A**	3GQA 081 301-••A	2840	75.0	75.5	0.85	1.7	6.1	2.52	2.2	3.0	0.00091	16	57
1.1	**M2QA 80 M2B**	3GQA 081 302-••A	2855	78.0	77.9	0.85	2.4	7.0	3.68	2.2	2.2	0.00107	17	58
1.5	**M2QA 90 S2A**	3GQA 091 101-••A	2850	79.0	79.0	0.87	3.15	7.0	5.03	2.2	2.5	0.00135	21	61
2.2	**M2QA 90 L2A**	3GQA 091 501-••A	2850	81.5	81.8	0.86	4.53	7.0	7.37	2.2	3.5	0.00163	24	61
3	**M2QA 100 L2A**	3GQA 101 501-••A	2860	83.0	83.2	0.88	5.93	7.0	10.02	2.2	3.0	0.00402	33	65
4	**M2QA 112 M2A**	3GQA 111 301-••A	2900	85.0	84.6	0.90	7.55	7.0	13.17	2.2	3.2	0.00671	42	67
5.5	**M2QA 132 S2A**	3GQA 131 101-••A	2920	87.5	87.9	0.89	10.2	7.0	17.99	2.2	3.0	0.01241	58	70
7.5	**M2QA 132 S2B**	3GQA 131 102-••A	2920	88.5	90.1	0.90	13.6	7.0	24.53	2.2	3.5	0.01491	63	70
11	**M2QA 160 M2A**	3GQA 161 301-••A	2930	90.0	90.5	0.89	19.8	6.5	35.85	2.5	3.1	0.0436	112	72
15	**M2QA 160 M2B**	3GQA 161 302-••A	2920	90.0	90.1	0.89	27.03	6.5	49.06	2.5	2.6	0.0551	122	72
18.5	**M2QA 160 L2A**	3GQA 161 501-••A	2930	90.5	90.9	0.90	32.8	6.5	60	2.5	2.7	0.06549	142	72
22	**M2QA 180 M2A**	3GQA 181 301-••A	2940	90.8	91.0	0.90	38.9	6.5	71	2.3	2.5	0.08805	170	72
30	**M2QA 200 L2A**	3GQA 201 501-••A	2955	91.4	91.1	0.90	52	6.5	96	2.2	2.6	0.14821	235	81
37	**M2QA 200 L2B**	3GQA 201 502-••A	2955	92.2	91.8	0.90	64	6.5	119	2.3	2.6	0.16822	254	81
45	**M2QA 225 M2A**	3GQA 221 301-••A	2970	92.6	92.2	0.89	78	7.0	144	2.5	2.7	0.29345	328	81
55	**M2QA 250 M2A**	3GQA 251 301-••A	2960	93.4	91.7	0.89	96	7.5	177	2.4	2.7	0.3784	390	84
75	**M2BAT 280 SMA**	3GBA 281 210-••D	2974	94.1	93.6	0.87	134	6.7	241	1.7	2.6	0.7	570	78
90	**M2BAT 280 SMB**	3GBA 281 220-••D	2970	94.5	94.2	0.89	156	6.4	289	1.7	2.5	0.82	610	78
110	**M2BAT 315 SMA**	3GBA 311 210-••D	2979	94.1	93.4	0.85	198	6.3	353	1.5	2.5	1.05	820	83
132	**M2BAT 315 SMB**	3GBA 311 220-••D	2977	94.7	94.1	0.87	232	6.3	423	1.7	2.5	1.25	870	83
160	**M2BAT 315 SMC**	3GBA 311 230-••D	2976	95.1	94.8	0.88	273	6.2	513	1.7	2.4	1.5	960	83
200	**M2BAT 315 MLA**	3GBA 311 410-••D	2980	95.7	95.3	0.88	345	7.9	641	2.6	3.1	1.95	1130	83
250	**M2BAT 355 S**	3GBA 351 100-••D	2983	95.7	95.3	0.89	424	6.8	800	1.5	2.8	2.7	1500	83
2-poles = 3000 r/min			**400 V 50 Hz**							**High-output design**				
5.5 [1]	**M2QA 112 L2 A**	3GQA 111 501-••A	2900	82.0	[2]	0.90	10.76	7.0	18.1	2.0	2.1	0.008263	49	70
9.2 [1]	**M2QA 132 M2A**	3GQA 131 301-••B	2910	85.5	[2]	0.88	17.65	7.5	30.2	2.0	2.2	0.014995	68	71
11 [1]	**M2QA 132 M2B**	3GQA 131 302-••B	2900	88.0	[2]	0.90	20.05	8.0	36.2	2.2	2.2	0.01768	73	73
22 [1]	**M2QA 160 L2B**	3GQA 161 502-••A	2930	88.0	[2]	0.90	40.09	6.5	71	2.3	2.8	0.06549	130	75
30 [1]	**M2QA 180 L2A**	3GQA 181 501-••A	2950	90.8	[2]	0.90	53	6.5	97	2.3	2.8	0.10339	185	75
45 [1]	**M2QA 200 L2C**	3GQA 201 503-••A	2955	92.0	[2]	0.90	78	7.0	145	2.2	2.6	0.18473	276	81
55 [1]	**M2QA 225 M2B**	3GQA 221 302-••A	2975	92.6	[2]	0.89	96	7.0	177	2.5	2.8	0.33431	340	81
75 [1]	**M2QA 250 M2B**	3GQA 251 302-••A	2970	91.0	[2]	0.89	134	7.0	241	2.4	2.8	0.45829	411	85
110	**M2BAT 280 SMC**	3GBA 281 230-••D	2973	95.0	94.8	0.90	187	6.7	353	1.9	2.6	1.05	660	78

[1] Temperature rise class F
[2] On request.

Efficiency classes fixed for ranges 1.1 to 90 kW
(available only by 2- and 4-poles).

General purpose cast iron motors

Technical data for totally enclosed squirrel cage three phase motors

IP 55, IC 411; Insulation class F, temperature rise class B

Output kW	Type designation	Product code	Speed r/min	Efficiency Full load 100%	Efficiency 3/4 load 75%	Power factor cos φ 100 %	Current I_N A	Current I_s/I_N	Torque T_N Nm	T_s/T_N	T_{max}/T_N	Moment of inertia J=1/4 GD² kgm²	Weight kg	Sound pressure level LP dB(A)
4-poles = 1500 r/min			**400 V 50 Hz**							**Basic design**			EFF 2	
0.25	**M2QA 71 M4A**	3GQA 072 301-••A	1395	65.5	63.3	0.72	0.77	5.2	1.71	2.1	2.7	0.00053	11	43
0.37	**M2QA 71 M4B**	3GQA 072 302-••A	1395	68.5	69.4	0.75	1.04	5.2	2.53	2.1	2.7	0.00066	11	45
0.55	**M2QA 80 M4A**	3GQA 082 301-••A	1410	73.5	71.4	0.72	1.5	5.2	3.73	2.4	2.7	0.00145	16	46
0.75	**M2QA 80 M4B**	3GQA 082 302-••A	1415	74.5	75.2	0.75	1.93	6.0	5.06	2.4	2.6	0.00174	17	46
1.1	**M2QA 90 S4A**	3GQA 092 101-••A	1400	77.5	77.8	0.78	2.65	6.0	7.5	2.3	2.4	0.00254	21	52
1.5	**M2QA 90 L4A**	3GQA 092 501-••A	1390	78.5	79.2	0.79	3.5	6.0	10.31	2.3	2.6	0.00317	25	52
2.2	**M2QA 100 L4A**	3GQA 102 501-••A	1430	81.5	82.3	0.81	4.85	6.0	14.69	2.3	2.7	0.00679	32	53
3	**M2QA 100 L4B**	3GQA 102 502-••A	1420	82.8	82.5	0.83	6.3	6.5	20.18	2.3	2.8	0.00862	36	53
4	**M2QA 112 M4A**	3GQA 112 301-••A	1430	85.0	84.6	0.82	8.29	6.5	26.71	2.3	2.8	0.01306	45	56
5.5	**M2QA 132 S4A**	3GQA 132 101-••A	1430	86.0	87.1	0.85	10.9	6.5	36.73	2.3	2.9	0.02673	60	59
7.5	**M2QA 132 M4A**	3GQA 132 301-••A	1440	88.5	88.3	0.85	14.4	6.5	49.74	2.3	2.7	0.03432	73	59
11	**M2QA 160 M4A**	3GQA 162 301-••A	1460	89.5	90.0	0.85	20.87	6.5	71	2.4	2.8	0.06543	116	66
15	**M2QA 160 L4A**	3GQA 162 501-••A	1460	90.0	90.4	0.86	27.97	6.5	98	2.3	2.4	0.09349	137	66
18.5	**M2QA 180 M4A**	3GQA 182 301-••A	1470	91.0	90.9	0.86	34.12	6.5	120	2.3	3.0	0.16049	170	66
22	**M2QA 180 L4A**	3GQA 182 501-••A	1470	91.5	90.0	0.88	39.44	6.5	142	2.4	3.0	0.18046	186	66
30	**M2QA 200 L4A**	3GQA 202 501-••A	1470	92.2	91.8	0.88	53	6.5	194	2.2	2.9	0.2819	254	71
37	**M2QA 225 S4A**	3GQA 222 101-••A	1480	92.6	91.2	0.85	67	7.0	238	2.2	2.7	0.37	308	73
45	**M2QA 225 M4A**	3GQA 222 301-••A	1480	92.8	91.7	0.87	80	7.0	290	2.2	2.7	0.42	335	73
55	**M2QA 250 M4A**	3GQA 252 301-••A	1480	93.4	91.3	0.87	98	7.0	354	2.4	2.7	0.78	450	76
75	**M2BAT 280 SMA**	3GBA 282 210-••D	1483	94.2	94.2	0.83	138	6.3	483	2.1	2.6	1.05	560	71
90	**M2BAT 280 SMB**	3GBA 282 220-••D	1481	94.6	94.7	0.86	162	6.4	580	2.1	2.4	1.32	600	71
110	**M2BAT 315 SMA**	3GBA 312 210-••D	1486	94.6	94.2	0.84	203	6.4	707	1.7	2.3	1.9	800	78
132	**M2BAT 315 SMB**	3GBA 312 220-••D	1485	94.9	94.7	0.85	239	6.1	849	1.9	2.4	2.2	855	78
160	**M2BAT 315 SMC**	3GBA 312 230-••D	1486	95.4	95.2	0.85	286	6.7	1028	2.1	2.6	2.6	930	78
200	**M2BAT 315 MLA**	3GBA 312 410-••D	1485	95.7	95.6	0.86	354	6.4	1286	2.1	2.5	3.2	1030	78
250	**M2BAT 355 S**	3GBA 352 100-••D	1488	95.6	95.3	0.85	448	6.7	1604	2.0	2.6	5.4	1500	82
4-poles = 1500 r/min			**400 V 50 Hz**							**High-output design**				
5.5 [1]	**M2QA 112 L4A**	3GQA 112 501-••A	1430	84.0	[2]	0.83	11.39	7.0	36.7	2.2	2.2	0.01484	49	64
9.2 [1]	**M2QA 132 M4B**	3GQA 132 302-••A	1430	84.0	[2]	0.85	18.6	6.5	61	2.2	2.2	0.0347	75	71
11 [1]	**M2QA 132 M4C**	3GQA 132 303-••A	1430	84.5	[2]	0.85	22.11	6.5	73	2.2	2.2	0.04227	80	73
18.5 [1]	**M2QA 160 L4B**	3GQA 162 502-••A	1460	87.0	[2]	0.86	35.69	6.5	121	2.2	2.4	0.10686	147	66
30 [1]	**M2QA 180 L4B**	3GQA 182 502-••A	1470	89.0	[2]	0.88	55	6.5	195	2.2	2.6	0.20783	200	70
37 [1]	**M2QA 200 L4B**	3GQA 202 502-••A	1470	89.2	[2]	0.88	68	6.5	240	2.2	2.6	0.29715	277	72
55 [1]	**M2QA 225 M4B**	3GQA 222 302-••A	1480	91.0	[2]	0.87	100	7.0	355	2.3	2.4	0.6244	351	75
75 [1]	**M2QA 250 M4B**	3GQA 252 302-••A	1480	90.4	[2]	0.87	137	7.0	484	2.3	2.4	0.9125	485	77
110	**M2BAT 280 SMC**	3GBA 282 230-••D	1484	95.1	95.1	0.85	196	7.1	708	2.7	2.8	1.7	660	71

[1] Temperature rise class F
[2] On request.

Efficiency classes fixed for ranges 1.1 to 90 kW
(available only by 2- and 4-poles).

General purpose cast iron motors

Technical data for totally enclosed squirrel cage three phase motors

IP 55, IC 411; Insulation class F, temperature rise class B

Output kW	Type designation	Product code	Speed r/min	Efficiency Full load 100%	Efficiency 3/4 load 75%	Power factor cos φ 100 %	Current I_N A	Current I_s/I_N	Torque T_N Nm	T_s/T_N	T_{max}/T_N	Moment of inertia J=1/4 GD² kgm²	Weight kg	Sound pressure level LP dB(A)
6-poles = 1000 r/min			**400 V 50 Hz**							**Basic design**				
0.18	**M2QA 71 M6A**	3GQA 073 301-••A	910	55.0	50.1	0.65	0.73	4.0	1.89	1.8	2.4	0.00056	10	42
0.25	**M2QA 71 M6B**	3GQA 073 302-••A	890	60.0	58.3	0.65	0.93	4.0	2.68	1.8	2.5	0.00074	11	42
0.37	**M2QA 80 M6A**	3GQA 083 301-••A	930	63.0	63.2	0.66	1.29	5.0	3.8	1.9	2.0	0.00159	17	45
0.55	**M2QA 80 M6B**	3GQA 083 302-••A	925	65.0	65.1	0.68	1.8	5.0	5.68	1.9	1.8	0.00196	18	45
0.75	**M2QA 90 S6A**	3GQA 093 101-••A	920	71.0	70.2	0.72	2.12	5.0	7.79	2.0	2.3	0.00292	21	48
1.1	**M2QA 90 L6A**	3GQA 093 501-••A	920	73.0	73.1	0.74	2.94	5.0	11.42	2.0	2.6	0.00379	25	48
1.5	**M2QA 100 L6A**	3GQA 103 501-••A	940	76.0	75.3	0.77	3.78	5.5	15.24	2.0	2.4	0.00999	32	51
2.2	**M2QA 112 M6A**	3GQA 113 301-••A	940	80.0	81.2	0.76	5.23	5.5	22.35	2.0	2.3	0.03116	40	54
3	**M2QA 132 S6A**	3GQA 133 101-••A	960	82.5	83.5	0.78	6.73	6.5	29.84	2.0	2.4	0.03116	55	56
4	**M2QA 132 M6A**	3GQA 133 301-••A	960	84.0	84.2	0.77	8.93	6.5	39.79	2.0	2.9	0.04074	65	56
5.5	**M2QA 132 M6B**	3GQA 133 302-••A	960	86.0	85.6	0.79	11.7	6.5	54	2.0	3.0	0.05332	75	56
7.5	**M2QA 160 M6A**	3GQA 163 301-••A	970	88.0	88.3	0.78	15.77	6.0	73	2.0	2.3	0.09231	119	61
11	**M2QA 160 L6A**	3GQA 163 501-••A	970	88.5	88.6	0.78	23	6.0	108	2.2	2.4	0.1297	140	62
15	**M2QA 180 L6A**	3GQA 183 501-••A	980	89.0	89.1	0.82	29.67	6.0	146	2.3	2.9	0.2418	180	63
18.5	**M2QA 200 L6A**	3GQA 203 501-••A	980	90.3	90.2	0.82	36.06	6.0	180	2.2	2.5	0.34174	231	64
22	**M2QA 200 L6B**	3GQA 203 502-••A	980	90.4	90.3	0.83	42.32	6.0	214	2.1	3.2	0.46837	254	64
30	**M2QA 225 M6A**	3GQA 223 301-••A	985	91.5	89.9	0.81	58	7.0	290	2.2	2.9	0.62691	308	66
37	**M2QA 250 M6A**	3GQA 253 301-••A	980	92.2	92.4	0.88	66	6.8	360	2.3	2.6	0.97	382	68
45	**M2BAT 280 SMA**	3GBA 283 210-••D	990	93.5	93.3	0.82	85	6.7	434	2.4	2.4	1.6	540	71
55	**M2BAT 280 SMB**	3GBA 283 220-••D	989	93.8	93.7	0.83	103	6.4	531	2.4	2.4	1.9	580	71
75	**M2BAT 315 SMA**	3GBA 313 210-••D	992	94.2	94.0	0.80	145	6.3	722	1.9	2.3	2.8	780	75
90	**M2BAT 315 SMB**	3GBA 313 220-••D	991	94.8	94.7	0.83	166	6.5	867	1.9	2.3	3.6	870	75
110	**M2BAT 315 SMC**	3GBA 313 230-••D	991	95.1	95.0	0.82	206	6.7	1060	2.1	2.6	4.4	930	75
132	**M2BAT 315 MLA**	3GBA 313 410-••D	991	95.3	95.2	0.83	242	6.5	1272	2.2	2.5	5.3	1040	75
160	**M2BAT 355 S**	3GBA 353 100-••D	992	95.6	95.4	0.83	293	6.2	1540	1.8	2.3	7.3	1500	77
6-poles = 1000 r/min			**400 V 50 Hz**							**High-output design**				
3 [1]	**M2QA 112 M6B**	3GQA 113 302-••A	950	77.0	[2]	0.76	740	6.5	30.2	1.9	2.1	0.0199	45	56
6.5 [1]	**M2QA 132 M6C**	3GQA 133 303-••A	970	83.0	[2]	0.78	14.49	6.5	64	1.9	2.1	0.0611	75	59
14 [1]	**M2QA 160 L6B**	3GQA 163 502-••A	970	85.5	[2]	0.78	30.3	6.0	138	2.1	2.2	0.139	155	64
18.5 [1]	**M2QA 180 L6B**	3GQA 183 502-••A	980	86.0	[2]	0.82	37.87	6.0	180	2.2	2.7	0.283984	196	65
30 [1]	**M2QA 200 L6C**	3GQA 203 503-••A	980	87.4	[2]	0.78	63	6.0	292	2.0	2.6	0.495	291	66
37 [1]	**M2QA 225 M6B**	3GQA 223 302-••A	980	87.8	[2]	0.78	78	6.6	361	2.1	2.6	0.803267	351	68
45 [1]	**M2QA 250 M6B**	3GQA 253 302-••A	980	89.2	[2]	0.88	82	6.8	439	2.2	2.6	1.32	455	71
75	**M2BAT 280 SMC**	3GBA 283 230-••D	989	94.5	94.5	0.83	139	6.9	724	2.6	2.5	2.6	660	71

[1] Temperature rise class F
[2] On request.

General purpose cast iron motors

Technical data for totally enclosed squirrel cage three phase motors

IP 55, IC 411; Insulation class F, temperature rise class B

Output kW	Type designation	Product code	Speed r/min	Efficiency Full load 100%	Efficiency 3/4 load 75%	Power factor cos φ 100 %	Current I_N A	Current I_s/I_N	Torque T_N Nm	T_s/T_N	T_{max}/T_N	Moment of inertia J=1/4 GD² kgm²	Weight kg	Sound pressure level LP dB(A)
8-poles = 750 r/min			**400 V 50 Hz**								**Basic design**			
0.18	**M2QA 80 M8A**	3GQA 084 301-••A	700	51.0	50.1	0.60	0.85	3.3	2.46	1.8	1.9	0.00111	16	42
0.25	**M2QA 80 M8B**	3GQA 084 302-••A	700	54.5	53.3	0.60	1.11	3.6	3.41	1.8	1.9	0.00326	17	42
0.37	**M2QA 90 S8A**	3GQA 094 101-••A	700	62.5	62.1	0.60	1.42	4.4	5.05	1.8	1.9	0.00541	21	46
0.55	**M2QA 90 L8A**	3GQA 094 501-••A	700	63.5	63.3	0.60	2.07	4.7	7.5	1.8	2.0	0.00756	24	46
0.75	**M2QA 100 L8A**	3GQA 104 501-••A	700	70.0	70.1	0.64	2.42	5.0	10.23	1.8	2.2	0.00971	31	53
1.1	**M2QA 100 L8B**	3GQA 104 502-••A	700	71.5	70.3	0.65	3.45	5.0	15.01	1.8	2.4	0.01186	34	53
1.5	**M2QA 112 M8A**	3GQA 114 301-••A	700	75.0	75.4	0.68	4.27	5.0	20.46	1.8	2.4	0.01559	42	55
2.2	**M2QA 132 S8A**	3GQA 134 101-••A	710	81.0	81.8	0.70	5.6	5.5	29.59	1.8	2.5	0.03625	56	55
3	**M2QA 132 M8A**	3GQA 134 301-••A	710	81.0	81.4	0.75	7.13	5.5	40.35	1.8	2.2	0.04141	64	56
4	**M2QA 160 M8A**	3GQA 164 301-••A	720	84.0	84.0	0.73	9.42	5.5	53	2.1	2.6	0.0676	105	58
5.5	**M2QA 160 M8B**	3GQA 164 302-••A	720	85.5	85.6	0.74	12.55	5.5	72	2.1	2.8	0.09524	125	58
7.5	**M2QA 160 L8A**	3GQA 164 501-••A	720	86.5	85.8	0.74	16.91	5.5	99	2.1	2.5	0.12122	142	58
11	**M2QA 180 L8A**	3GQA 184 501-••A	730	87.7	87.0	0.77	23.51	5.4	143	2.0	2.8	0.23645	176	61
15	**M2QA 200 L8A**	3GQA 204 501-••A	730	89.0	89.4	0.76	32.009	5.5	196	2.3	2.8	0.37103	235	63
18.5	**M2QA 225 S8A**	3GQA 224 101-••A	740	90.0	89.1	0.75	39.56	5.5	238	2.1	2.7	0.53287	290	65
22	**M2QA 225 M8A**	3GQA 224 301-••A	740	90.5	88.2	0.75	46.78	6.0	283	2.2	2.7	0.65825	302	65
30	**M2QA 250 M8A**	3GQA 254 301-••A	740	91.3	90.1	0.79	60	6.5	387	2.3	2.4	0.975	392	67
37	**M2BAT 280 SMA**	3GBA 284 210-••D	741	93.4	93.3	0.78	74	7.3	477	1.7	3.0	1.85	570	65
45	**M2BAT 280 SMB**	3GBA 284 220-••D	741	94.0	93.8	0.78	90	7.6	580	1.9	3.2	2.2	610	65
55	**M2BAT 315 SMA**	3GBA 314 210-••D	742	94.1	94.0	0.81	104	7.1	708	1.6	2.7	3.2	820	65
75	**M2BAT 315 SMB**	3GBA 314 220-••D	741	94.5	94.4	0.82	140	7.1	968	1.7	2.7	4.1	910	65
90	**M2BAT 315 SMC**	3GBA 314 230-••D	741	94.8	94.7	0.82	167	7.4	1161	1.8	2.7	4.9	980	65
110	**M2BAT 315 MLA**	3GBA 314 410-••D	740	95.0	95.0	0.83	202	7.3	1420	1.8	2.7	5.8	1100	72
132	**M2BAT 355 S**	3GBA 354 100-••D	743	95.2	95.1	0.81	247	6.5	1697	1.3	2.3	7.3	1500	75
8-poles = 750 r/min			**400 V 50 Hz**								**High-output design**			
2 [1]	**M2QA 112 M8B**	3GQA 114 302-••A	700	72.0	[2]	0.68	5.94	5.2	27.3	1.7	1.9	0.0199	45	58
3.8 [1]	**M2QA 132 M8B**	3GQA 134 302-••A	710	78.0	[2]	0.75	9.38	5.5	51	1.7	1.9	0.04776	75	59
8.5 [1]	**M2QA 160 L8B**	3GQA 164 502-••A	720	83.5	[2]	0.74	19.86	5.5	113	2.0	2.4	0.1312	136	61
15 [1]	**M2QA 180 L8B**	3GQA 184 502-••A	730	84.7	[2]	0.77	33.2	5.4	196	1.9	2.6	0.283984	196	63
18.5 [1]	**M2QA 200 L8B**	3GQA 204 502-••A	730	86.0	[2]	0.76	40.85	5.4	242	1.9	2.6	0.46854	274	65
30 [1]	**M2QA 225 M8B**	3GQA 224 302-••A	740	87.5	[2]	0.75	66	6.3	387	2.1	2.6	0.803267	349	67
37 [1]	**M2QA 250 M8B**	3GQA 254 302-••A	740	88.3	[2]	0.79	76	6.5	478	2.2	2.5	1.28	436	69
55	**M2BAT 280 SMC**	3GBA 284 230-••D	741	94.4	94.3	0.79	108	7.8	709	1.9	3.2	2.85	690	65

[1] Temperature rise class F
[2] On request.

General purpose motors - Variant codes

Code	Variant	Aluminum motors					Steel, cast iron motors			
		56-80	90-100	112-132	160-180	200-280	71-132	160-250	280-315	355-400
Bearings and Lubrication										
037	Roller bearing at D-end. Transport lock included.	NA	M	NA	M	M	NA	M	M	M/R
039	Cold resistant grease.	M	M	M	M	M	M	M	M/NA	M/NA
040	Heat resistant grease.	M	M	S	S	S	M	M	M	M/P
041	Bearings regreasable via grease nipples.	NA	M	M	M	S/M	NA/M	M	S	S
043	SPM nipples.	NA	R	M	M	M	NA/M	M	M	M/P
057	2RS bearings at both ends.	M	M	M	M/R	M/R	NA	NA	NA	NA
Branch standard design										
178	Stainless steel/acid proof bolts.	M	M	M	M	M	M	M	M/P	M/P
Cooling system										
053	Metal fan cover.	S	S	M	S	S	S	S	S	S
068	Metal fan.	NA/M	M	M	M	M	M	M	M/P	M/P
075	Cooling method IC418 (without fan).	P	P	R	R	R	NA	NA	P/NA	P/NA
183	Separate motor cooling (fan axial, N-end).	NA/M	R	NA	M/R	M/R	P/M	M	NA	NA
Earthing bolt										
067	External earthing bolt.	M	M	M	M	M	M	M	M	S/P
Heating elements										
450	Heating element 100-120 V.	M	M	M	M	M	M	M	M/P	M/P
451	Heating element 200-240 V.	M	M	M	M	M	M	M	M	M/P
Mounting arrangements										
007	IM 3001 flange mounted, from IM 1001 (B5 from B3).	NA/M	M	NA	NA	M	NA	NA	NA	NA
008	IM 2101 foot/flange mounted, from IM 1001 (B34 from B3).	NA/M	M	M	M	NA	M	M/NA	NA	NA
009	IM 2001 foot/flange mounted, from IM 1001 (B35 from B3).	M	M	M	M	M	M	M	M	M/P
047	IM 3601 flange mounted, from IM 3001 (B14 from B5).	M	M	M	M	NA	M	M/NA	NA	NA
048	IM 3001 flange mounted, from IM 3601 (B5 from B14).	M	M	M	NA	NA	NA	NA	NA	NA
066	Modified for non-standard mounting position. (Please specify IMxxxx.) Must be ordered for all mounting arrangements excluding IM B3 (1001) and B5 (3001).	M	M	M	M	M	M	NA	M/NA	M/P
Painting										
114	Special paint colour, standard grade.	M	M	M	M	M	M	M	M	M
179	Special paint specification.	R	R	R	R	R	NA	NA	R/NA	R/NA
Protection										
005	Protective roof, vertical motor, shaft down. Vertically mounted motors with shaft extension downwards.	M	M	M	M	M	M	M	M	M/P
072	Radial seal at D-end.	M	M	M/R	M/R	M/R	M	M	M	MP
158	Degree of protection IP65 or IP65X. Dust proof version.	NA	P	M	M	M	M	M	NA	NA
211	Weather protected, IP xx W.	NA	P	P	P	P	M	M	NA	NA
403	Degree of protection IP56.	M	P	M	M	M	M	M	M	M/P
Rating & instruction plates										
002	Restamping voltage, frequency and output, continuous duty.	M	M	M	M	M	M	M	M	M/P
003	Individual serial number.	P	M	M	M	M	S	S	S	S
098	Stainless rating plate.	M	M	M	M/R	M/R	S	S	S	S
138	Mounting of additional identification plate, aluminum.	M	M	M	M	M	M	M	M	M/P
Stator winding temperature sensors										
122	Bimetal detectors, break type (NCC), (3 in series), 150°C.	M	M	M	M	M	M	M	M	M/P
436	PTC - thermistors (3 in series), 150 °C.	M	M	M	M	S/M	S	S	S	S
439	PTC - thermistors (2x3 in series), 150 °C.	M	M	M	M	M	M	M	M	M/P
445	Pt100 (1 per phase).	NA	R	M	M	M	M	M	M	M/P
Terminal box										
021	Terminal box left-hand side, seen from D-end.	NA/M	M	NA	NA	P/NA	P	P	P	P/NA
180	Terminal box right-hand side, seen from D-end.	NA/M	M	NA	NA	P/R	P	P	P	P
230	Standard gable glands.	M	M	M	M	M	M	M	S/NA	S/NA
731	Two standard cable glands.	M	M	M	M	M	M	M	NA/S	NA/S
Testing										
145	Type test report from test of identical motor. 400V 50 Hz.	M	M	M	M	M	M	M	M	M/P
148	Routine test report.	P	M	M	M	M	M	M	M	M/P

Certain variant codes cannot be used together.

S = Included as standard.
M = On modification of a stocked motor, or on new manufacture,the number per order may be limited.
P = New manufacture only.
R = On request.
NA = Not applicable.

Note! This list is an extract of a wide range of possible modifications. Please see the General Purpose or Process Performance Motor Catalogue for a complete list of variant codes.

Motors in brief, Basic design

Size		56	63	71	80	90	100	112	132	160	180	200	225	250	280	315	355	400
General purpose motors, aluminum																		
Bearings	D-end	6201-2Z/C3	6202-2Z/C3	6203-2Z/C3	6204-2Z/C3	6205-2Z/C3	6306-2Z/C3	6206-2Z/C3	6208-2Z/C3	6309-2Z/C3	6310-2Z/C3	6312/C3	6313/C3	6315/C3	6316/C3 [1]			
	N-end [2]	6201-2Z/C3	6201-2Z/C3	6202-2Z/C3	6203-2Z/C3	6204-2Z/C3	6205-2Z/C3	6205-2Z/C3	6206-2Z/C3	6209-2Z/C3	6209-2Z/C3	6210/C3	6212/C3	6213/C3	6213/C3			
	N-end [3]							6206-2Z/C3	6208-2Z/C3									
	[1] 6315/C3 for 2-pole motors [2] M3AA 112: M-6, M-8, M3AA 132: SA-2, S-4, S-6, MA-6, MB-6, S-8 and M-8. M2AA 200: 6209-2Z/C3; M2AA 225: 6210/C3; M2AA 250: 6212/C3 [3] All M3AA motor sizes 112-132 not included in [2].																	
Axially-locked bearings	Inner bearing cover	Spring washer at N-end.				D-end		D-end [1]		D-end								
								[1] Foot motor. A spring washer at the N-end presses the motor towards the D-end. Flange motor. Inner bearing cover and spring-washer at the N-end.										
Lubrication		Permanently lubricated bearings.						Permanently lubricated shielded bearings (incl. M2AA 200).				Valve lubrication.						
Connections	Cable entries	1xM16xPg11		2xM20xPg16		4xM25		4x(M25+M20)		2x(2xM40+M16)* *incl. M2AA 200		1x(2xM40+M16)			1xM63+M16			
	Screw	M4						M5		M6		M10						
	Terminal box	Screw terminal, 6 terminals.						6 terminals for connection with cable lugs (not included).										
Winding protection		Optional.										PTC thermistors, 150°C, 3 in series (optional for M2AA 200).						
Drain holes		Standard at D-end.		Standard at both ends.														
General purpose motors, cast iron																		
Bearings	D-end, 2 pole			6202	6204	6205	6206	6207	6208	6309	6310	6312	6313	6314/C3	6316/C4	6316/C4	6319/C3	
	D-end, 4-8 pole			DDU C3	DDU C3	DDU C3	DDU C3	DDU C3	DDU C3	DDU C3	DDU C3	DDU C3	ZZ C3		6316/C3	6319/C3	6322/C3	
	N-end, 2 pole			6202	6204	6205	6206	6206	6207	6209	6210	6212	6213	6214/C3	6316/C4	6316/C4	6319/C3	
	N-end, 4-8 pole			DDU C3	DDU C3	DDU C3	DDU C3	DDU C3	DDU C3	DDU C3	DDU C3	DDU C3	ZZ C3		6316/C3	6316/C3	6319/C3	
Axially-locked bearings	Inner bearing cover			As standard, locked at D-end.														
Lubrication				Greased for life							Greased for life or regreasable.				Regreasable bearings.			
Connections	Cable entries			2xM16x1.5	2xM25x1.5	2xM32x1.5				2xM40x1.5		2xM50x1.5		2xM63x1.5	2xM63+2xM20			
	Screw														M8	M10	M12/M10	
	Terminal box			6 terminals for connection with cable lugs (not included).														
Winding protection				Optional.											3 PTC thermistors as standard, 150 °C.			
Drain holes				Optional.											Standard			
General purpose motors, steel																		
Bearings	D-end, 2 pole														6316/C4	6316/C4	6316M/C3	6317M/C3
	D-end, 4-8 pole														6316/C3	6319/C3	6322/C3	6322/C3
	N-end, 2 pole														6316/C4	6316/C4	6316M/C3	6317M/C3
	N-end, 4-8 pole														6316/C3	6316/C3	6319/C3	6319/ C3
Axially-locked bearings	Inner bearing cover														As standard, locked at D-end.			
Lubrication															Regreasing nipples, M10x1.			
SPM-nippless															On request.			
Connections	Cable entries														2xM63+2xM20			
	Screw														M12			
	Terminal box														6 terminals for connection with cable lugs (not incl.).			
Winding protection															PTC thermistors 150 °C, 3 in series, as standard.			
Drain holes															Standard.			

Process performance cast iron and aluminum motors

Technical data for totally enclosed squirrel cage three phase motors

IP 55, IC 411; Insulation class F, temperature rise class B

Output kW		Type designation	Product code	Speed r/min	Efficiency Full load 100%	Efficiency 3/4 load 75%	Power factor cos φ 100 %	Current I_N A	Current I_s/I_N	Torque T_N Nm	Torque T_s/T_N	Torque T_{max}/T_N	Moment of inertia J=1/4 GD² kgm²	Weight kg	Sound pressure level LP dB(A)
2-poles = 3000 r/min				**400 V 50 Hz**							**Basic design**			EFF I	
0.37		**M2BA 71 M2 A**	3GBA 071 310-••A	2810	71.0	68.1	0.80	0.94	6.1	1.26	2.2	3.0	0.0003	10	56
0.55		**M2BA 71 M2 B**	3GBA 071 320-••A	2800	74.0	71.4	0.82	1.31	6.1	1.88	2.2	2.7	0.0004	11	56
0.75		**M2BA 80 M2 A**	3GBA 081 310-••A	2850	77.2	75.5	0.86	1.63	6.1	2.51	2.2	3.0	0.0009	16	57
1.1	2)	**M2BA 80 M2 B**	3GBA 081 320-••A	2850	80.2	77.6	0.85	2.33	7.0	3.69	2.2	2.2	0.0011	17	58
1.5	2)	**M2BA 90 S2 A**	3GBA 091 110-••A	2850	81.6	79.0	0.85	3.13	7.0	5.03	2.2	2.5	0.0014	21	61
2.2	2)	**M2BA 90 L2 A**	3GBA 091 510-••A	2850	84.2	81.9	0.84	4.49	7.0	7.37	2.2	3.5	0.0016	24	61
3	2)	**M2BA 100 L2 A**	3GBA 101 510-••A	2870	85.1	83.2	0.86	5.92	7.0	9.98	2.2	3.0	0.004	33	65
4	2)	**M2BA 112 M2 A**	3GBA 111 310-••A	2900	86.0	84.5	0.89	7.52	7.0	13.17	2.2	3.2	0.0067	42	67
5.5		**M2BA 132 S2 A**	3GBA 131 110-••A	2920	88.6	88.1	0.88	10.19	7.0	17.99	2.2	3.0	0.0124	58	70
7.5		**M2BA 132 S2 B**	3GBA 131 120-••A	2920	89.9	88.7	0.89	13.54	7.0	24.53	2.2	3.4	0.0149	63	70
11		**M3BP 160 MA**	3GBP 161 101-••A	2930	91.0	91.2	0.88	20	6.2	36	2.1	2.8	0.039	105	69
15		**M3BP 160 M**	3GBP 161 102-••A	2920	91.3	91.7	0.90	26.5	6.4	49	2.3	2.7	0.047	118	69
18.5		**M3BP 160 L**	3GBP 161 103-••A	2920	92.4	93.1	0.91	32	7.2	61	2.6	2.9	0.053	133	69
22		**M3BP 180 M**	3GBP 181 101-••A	2930	92.8	93.3	0.89	38.5	7.2	71	2.7	3.0	0.077	178	69
30		**M3BP 200 MLA**	3GBP 201 001-••A	2955	93.2	93.2	0.88	53	8.5	97	2.9	3.1	0.15	250	72
37		**M3BP 200 MLB**	3GBP 201 002-••A	2950	93.6	93.7	0.89	64	7.2	120	2.3	2.9	0.18	270	72
45		**M3BP 225 SMB**	3GBP 221 001-••A	2960	94.1	93.9	0.88	79	7.7	145	2.5	2.9	0.26	335	74
55		**M3BP 250 SMA**	3GBP 251 001-••A	2970	94.2	93.8	0.89	95	7.9	177	2.4	3.0	0.49	420	75
75	3)	**M3BP 280 SMA**	3GBP 281 210-••G	2978	94.8	94.3	0.88	131	7.6	240	2.1	3.0	0.8	625	77
90	3)	**M3BP 280 SMB**	3GBP 281 220-••G	2976	95.1	94.8	0.90	152	7.4	289	2.1	2.9	0.9	665	77
110	3)	**M3BP 315 SMA**	3GBP 311 210-••G	2982	95.1	94.4	0.86	194	7.6	352	2.0	3.0	1.2	880	78
132	3)	**M3BP 315 SMB**	3GBP 311 220-••G	2982	95.5	95.0	0.88	228	7.4	423	2.2	3.0	1.4	940	78
160	3)	**M3BP 315 SMC**	3GBP 311 230-••G	2981	96.1	95.6	0.89	269	7.5	513	2.3	3.0	1.7	1025	78
200	3)	**M3BP 315 MLA**	3GBP 311 410-••G	2980	96.3	95.9	0.90	336	7.7	641	2.6	3.0	2.1	1190	78
250	3)	**M3BP 355 SMA**	3GBP 351 210-••G	2984	96.4	95.9	0.89	425	7.7	800	2.1	3.3	3	1600	83
315	3)	**M3BP 355 SMB**	3GBP 351 220-••G	2980	96.6	96.3	0.89	535	7.0	1009	2.1	3.0	3.4	1680	83
355	3)	**M3BP 355 SMC**	3GBP 351 230-••G	2984	96.8	96.5	0.88	604	7.2	1136	2.2	3.0	3.6	1750	83
400	3)	**M3BP 355 MLA**	3GBP 351 410-••G	2982	96.9	96.7	0.88	680	7.1	1281	2.3	2.9	4.1	2000	83
450	3)	**M3BP 355 MLB**	3GBP 351 420-••G	2983	97.1	97.0	0.90	750	7.9	1441	2.2	2.9	4.3	2080	83
500	3)	**M3BP 355 LKA**	3GBP 351 810-••G	2982	97.1	97.0	0.90	830	7.5	1601	2.1	3.5	4.8	2320	83
560	3)	**M3BP 355 LKB**	3GBP 351 820-••G	2982	97.2	97.1	0.90	930	8.0	1793	2.3	3.6	5.2	2460	83
560	4)	**M3BP 400 LA**	3GBP 401 510-••G	2988	97.2	97.0	0.89	940	7.8	1790	2.1	3.4	7.9	2950	82
560	4)	**M3BP 400 LKA**	3GBP 401 810-••G	2988	97.2	97.0	0.89	940	7.8	1790	2.1	3.4	7.9	2950	82
630	4)	**M3BP 400 LB**	3GBP 401 520-••G	2987	97.4	97.3	0.89	1055	7.8	2014	2.2	3.4	8.2	3050	82
630	4)	**M3BP 400 LKB**	3GBP 401 820-••G	2987	97.4	97.3	0.89	1055	7.8	2014	2.2	3.4	8.2	3050	82
710	4)	**M3BP 400 LC**	3GBP 401 530-••G	2987	97.5	97.4	0.89	1185	7.8	2270	2.6	3.4	9.3	3300	82
710	4)	**M3BP 400 LKC**	3GBP 401 830-••G	2987	97.5	97.4	0.89	1185	7.8	2270	2.6	3.4	9.3	3300	82
800	4)	**M3BP 450 LA**	3GBP 451 510-••G	2990	97.3	97.2	0.88	1345	7.8	2555	1.3	3.2	12.5	4000	85
900	4)	**M3BP 450 LB**	3GBP 451 520-••G	2990	97.4	97.3	0.88	1515	7.8	2874	1.5	3.1	14	4200	85
1000	1) 4)	**M3BP 450 LC**	3GBP 451 530-••G	2990	97.6	97.5	0.89	965	7.8	3194	1.6	3.2	15.5	4400	85
4	5)	**M3AP 112 M**	3GAA 111 022-••C	2860	87.7	89.4	0.93	7.1	7.5	13.4	2.7	3.1	0.012	33	63
5.5	5)	**M3AP 132 SA**	3GAA 131 023-••C	2900	88.6	89.6	0.88	10.3	9.7	18.1	3.8	4.3	0.016	42	69
7.5	5)	**M3AP 132 SB**	3GAA 131 024-••C	2915	90.9	91.3	0.90	13.3	11.0	24.6	5.1	5.2	0.022	56	69
11	5)	**M3AP 160 MA**	3GAA 161 101-••C	2930	91.0	91.2	0.88	20	6.2	36	2.1	2.8	0.039	81	69
15	5)	**M3AP 160 M**	3GAA 161 102-••C	2920	91.3	91.7	0.90	26.5	6.4	49	2.3	2.7	0.047	92	69
18.5	5)	**M3AP 160 L**	3GAA 161 103-••C	2920	92.4	93.1	0.91	32	7.2	61	2.6	2.9	0.053	102	69
22	5)	**M3AP 180 M**	3GAA 181 101-••C	2930	92.8	93.3	0.89	38.5	7.2	71	2.7	3.0	0.077	128	69
30	5)	**M3AP 200 MLA**	3GAA 201 001-••C	2955	93.2	93.2	0.88	53	8.5	97	2.9	3.1	0.15	192	72
37	5)	**M3AP 200 MLB**	3GAA 201 002-••C	2950	93.6	93.7	0.89	64	7.2	120	2.3	2.9	0.18	217	72
45	5)	**M3AP 225 SMB**	3GAA 221 001-••C	2960	94.1	93.9	0.88	79	7.7	145	2.5	2.9	0.26	257	74
55	5)	**M3AP 250 SMA**	3GAA 251 001-••C	2970	94.2	93.8	0.89	95	7.9	177	2.4	3.0	0.49	311	75
75	5)	**M3AP 280 SMA**	3GAA 281 001-••C	2970	95.1	95.2	0.90	127	8.2	241	2.4	3.1	0.57	350	75
90	1) 5)	**M3AP 280 SMB**	3GAA 281 002-••C	2970	95.4	94.8	0.90	152	8.3	290	2.7	3.4	0.59	404	75
2-poles = 3000 r/min				**400 V 50 Hz**							**High-output design**				
22	1)	**M3BP 160 LB**	3GBP 161 104-••A	2920	92.0	93.0	0.91	38	6.9	72	2.3	2.9	0.058	140	69
30		**M3BP 180 LB**	3GBP 181 102-••A	2945	93.7	94.0	0.89	53	7.8	97	2.7	3.0	0.092	194	70
45		**M3BP 200 MLC**	3GBP 201 003-••A	2950	94.1	94.5	0.89	78	8.2	146	3.0	3.2	0.19	280	72
55		**M3BP 225 SMC**	3GBP 221 002-••A	2960	94.5	94.6	0.89	95	7.3	177	2.8	3.0	0.29	355	74
75	1)	**M3BP 250 SMB**	3GBP 251 002-••A	2970	95.0	94.9	0.90	127	8.6	241	2.7	3.3	0.57	465	75
110	3)	**M3BP 280 SMC**	3GBP 281 230-••G	2978	95.7	95.3	0.90	185	7.9	353	2.4	3.0	1.15	725	77
250	3)	**M3BP 315 LKA**	3GBP 311 810-••G	2980	96.5	96.4	0.89	422	8.1	801	2.8	2.9	2.65	1440	78
315	1) 3)	**M3BP 315 LKC**	3GBP 311 830-••G	2981	96.7	96.6	0.89	530	8.8	1009	3.2	3.2	3.3	1630	78
5.5	1) 5)	**M3AP 112 MB**	3GAA 111 002-••C	2855	86.6	87.9	0.93	9.9	7.3	18.4	2.6	3.4	0.012	33	63
9.2	1) 5)	**M3AP 132 SBB**	3GAA 131 004-••C	2840	86.8	88.3	0.92	16.8	8.5	31	3.3	3.6	0.02	50	69
11	1) 5)	**M3AP 132 SC**	3GAA 131 003-••C	2835	87.9	89.2	0.93	19.6	7.7	37	3.0	3.2	0.022	59	69
22	1) 5)	**M3AP 160 LB**	3GAA 161 104-••C	2920	92.0	93.0	0.91	38	6.9	72	2.3	2.9	0.058	108	69
30	5)	**M3AP 180 LB**	3GAA 181 102-••C	2945	93.7	94.0	0.89	53	7.8	97	2.7	3.0	0.092	146	70
45	5)	**M3AP 200 MLC**	3GAA 201 003-••C	2950	94.1	94.5	0.89	78	8.2	146	3.0	3.2	0.19	222	72
55	1) 5)	**M3AP 200 MLD**	3GAA 201 004-••C	2940	94.0	94.4	0.89	95	7.9	179	3.1	3.1	0.2	232	0
55	5)	**M3AP 225 SMC**	3GAA 221 002-••C	2960	94.5	94.6	0.89	95	7.3	177	2.8	3.0	0.29	282	74
80	1) 5)	**M3AP 225 SMD**	3GAA 221 003-••C	2960	94.7	94.7	0.86	143	7.5	258	2.9	3.1	0.3	275	74
75	5)	**M3AP 250 SMB**	3GAA 251 002-••C	2970	95.0	94.9	0.90	127	8.6	241	2.7	3.3	0.57	395	75
95	1) 5)	**M3AP 250 SMC**	3GAA 251 003-••C	2965	95.4	95.6	0.90	160	8.0	306	2.6	3.1	0.59	395	75

1) Temperature rise class F

2) Efficiency class 2.

3) -3dB(A) sound pressure level reduction with unidirectional fan construction. Direction of rotation must be stated when ordering, see variant codes 044 and 045.

4) Unidirectional fan construction as standard. Direction of rotation must be stated when ordering, see variant codes 044 and 045.

5) When ordering, the following variant code has to be added to the product code: 199 = Extreme heavy duty design. Type designation M3AP.

Efficiency classes fixed for ranges 1.1 to 90 kW (available only by 2- and 4-poles).

Process performance cast iron and aluminum motors

Technical data for totally enclosed squirrel cage three phase motors

IP 55, IC 411; Insulation class F, temperature rise class B

Output kW		Type designation	Product code	Speed r/min	Efficiency Full load 100%	Efficiency 3/4 load 75%	Power factor cos φ 100 %	Current I_N A	Current I_s/I_N	Torque T_N Nm	Torque T_s/T_N	Torque T_{max}/T_N	Moment of inertia J=1/4 GD² kgm²	Weight kg	Sound pressure level LP dB(A)
4-poles = 1500 r/min				**400 V 50 Hz**									**Basic design**	EFF I	
0.25		**M2BA 71 M4 A**	3GBA 072 310-••A	1390	66.3	63.3	0.73	0.75	5.2	1.72	2.1	2.7	0.0005	11	43
0.37		**M2BA 71 M4 B**	3GBA 072 320-••A	1380	70.8	69.4	0.75	1.01	5.2	2.56	2.1	2.6	0.0007	11	45
0.55		**M2BA 80 M4 A**	3GBA 082 310-••A	1410	75.0	72.4	0.73	1.45	5.2	3.73	2.4	2.7	0.0014	16	46
0.75		**M2BA 80 M4 B**	3GBA 082 320-••A	1400	76.3	75.1	0.76	1.87	6.0	5.12	2.4	2.6	0.0017	17	46
1.1	2)	**M2BA 90 S4 A**	3GBA 092 110-••A	1400	78.5	77.8	0.78	2.6	6.0	7.5	2.3	2.4	0.0025	21	52
1.5	2)	**M2BA 90 L4 A**	3GBA 092 510-••A	1390	80.5	79.2	0.78	3.45	6.0	10.31	2.3	2.6	0.0037	26	52
2.2	2)	**M2BA 100 L4 A**	3GBA 102 510-••A	1430	82.5	81.7	0.80	4.82	6.0	14.69	2.3	2.7	0.0068	32	53
3	2)	**M2BA 100 L4 B**	3GBA 102 520-••A	1420	84.5	82.5	0.82	6.25	6.5	20.18	2.3	2.8	0.0086	36	53
4	2)	**M2BA 112 M4 A**	3GBA 112 310-••A	1430	86.0	84.7	0.81	8.24	6.5	26.71	2.3	2.8	0.0131	45	56
5.5	2)	**M2BA 132 S4 A**	3GBA 132 110-••A	1430	87.4	87.1	0.84	10.82	6.5	36.73	2.3	2.9	0.0267	60	59
7.5	2)	**M2BA 132 M4 A**	3GBA 132 310-••A	1440	89.0	88.7	0.85	14.34	6.5	49.74	2.3	2.7	0.0343	73	59
11		**M3BP 160 M**	3GBP 162 101-••A	1460	92.0	92.7	0.81	21.5	7.6	72	2.9	3.4	0.091	115	62
15		**M3BP 160 L**	3GBP 162 102-••A	1460	91.8	92.5	0.82	29	8.1	98	3.3	3.6	0.102	135	62
18.5		**M3BP 180 M**	3GBP 182 101-••A	1470	92.3	92.9	0.84	35	7.0	120	2.9	2.9	0.161	175	62
22		**M3BP 180 L**	3GBP 182 102-••A	1470	93.1	93.9	0.85	40	7.0	143	2.6	2.8	0.225	203	63
30		**M3BP 200 MLB**	3GBP 202 001-••A	1475	93.4	94.0	0.84	55	7.5	194	2.5	2.8	0.34	275	63
37		**M3BP 225 SMA**	3GBP 222 001-••A	1480	93.6	93.8	0.84	68	7.4	239	2.9	3.0	0.37	310	66
45		**M3BP 225 SMB**	3GBP 222 002-••A	1480	94.2	94.4	0.83	83	7.6	291	2.8	3.0	0.42	330	66
55		**M3BP 250 SMA**	3GBP 252 001-••A	1480	94.6	94.9	0.86	98	7.6	355	3.1	3.0	0.72	420	67
75		**M3BP 280 SMA**	3GBP 282 210-••G	1484	94.9	94.8	0.85	135	6.9	483	2.5	2.8	1.25	625	68
90		**M3BP 280 SMB**	3GBP 282 220-••G	1483	95.3	95.3	0.86	159	7.2	580	2.5	2.7	1.5	665	68
110		**M3BP 315 SMA**	3GBP 312 210-••G	1487	95.6	95.4	0.86	193	7.2	706	2.0	2.5	2.3	900	70
132		**M3BP 315 SMB**	3GBP 312 220-••G	1487	95.8	95.7	0.86	232	7.1	848	2.3	2.7	2.6	960	70
160		**M3BP 315 SMC**	3GBP 312 230-••G	1487	96.0	95.9	0.85	287	7.2	1028	2.4	2.9	2.9	1000	70
200		**M3BP 315 MLA**	3GBP 312 410-••G	1486	96.2	96.2	0.86	351	7.2	1285	2.5	2.9	3.5	1160	70
250		**M3BP 355 SMA**	3GBP 352 210-••G	1488	96.5	96.4	0.86	438	7.1	1604	2.3	2.7	5.9	1610	74
315		**M3BP 355 SMB**	3GBP 352 220-••G	1488	96.7	96.6	0.86	550	7.3	2022	2.3	2.8	6.9	1780	74
355		**M3BP 355 SMC**	3GBP 352 230-••G	1487	96.7	96.6	0.86	616	6.8	2280	2.4	2.7	7.2	1820	78
400		**M3BP 355 MLA**	3GBP 352 410-••G	1489	96.9	96.7	0.85	700	6.8	2565	2.3	2.6	8.4	2140	78
450		**M3BP 355 MLB**	3GBP 352 420-••G	1490	96.9	96.7	0.86	784	6.9	2884	2.3	2.9	8.4	2140	78
500		**M3BP 355 LKA**	3GBP 352 810-••G	1490	97.0	96.9	0.86	875	6.8	3204	2.0	3.0	10	2500	78
560	1)	**M3BP 355 LKB**	3GBP 352 820-••G	1490	96.9	96.9	0.85	990	7.2	3589	2.6	2.7	10.6	2600	78
560		**M3BP 400 LA**	3GBP 402 510-••G	1491	97.1	97.0	0.85	980	7.4	3587	2.4	3.0	15	3200	78
560		**M3BP 400 LKA**	3GBP 402 810-••G	1491	97.1	97.0	0.85	980	7.4	3587	2.4	3.0	15	3200	78
630		**M3BP 400 LB**	3GBP 402 520-••G	1491	97.1	97.0	0.87	1085	7.6	4035	2.2	3.1	16	3300	78
630		**M3BP 400 LKB**	3GBP 402 820-••G	1491	97.1	97.0	0.87	1085	7.6	4035	2.2	3.1	16	3300	78
710	1)	**M3BP 400 LC**	3GBP 402 530-••G	1491	97.2	97.1	0.86	1240	7.6	4547	2.4	3.2	17	3400	78
710	1)	**M3BP 400 LKC**	3GBP 402 830-••G	1491	97.2	97.1	0.86	1240	7.6	4547	2.4	3.2	17	3400	78
800		**M3BP 450 LA**	3GBP 452 510-••G	1492	97.0	96.9	0.86	1385	7.0	5120	1.3	2.8	23	4050	85
900		**M3BP 450 LB**	3GBP 452 520-••G	1492	97.1	97.0	0.86	1555	7.0	5760	1.3	2.8	25	4350	85
1000	1)	**M3BP 450 LC**	3GBP 452 530-••G	1491	97.2	97.1	0.86	1725	6.8	6405	1.3	2.7	30	4700	85
3	3)	**M3AP 112 MA**	3GAA 112 021-••C	1455	87.5	87.8	0.81	6.2	7.9	19.7	2.7	3.7	0.018	34	56
4	3)	**M3AP 112 M**	3GAA 112 022-••C	1455	89.3	89.6	0.76	8.6	8.5	26.3	3.3	4.3	0.018	34	56
5.5	3)	**M3AP 132 S**	3GAA 132 023-••C	1460	89.3	90.5	0.84	10.6	7.5	36	2.6	3.1	0.038	48	59
7.5	3)	**M3AP 132 M**	3GAA 132 024-••C	1450	90.1	91.4	0.87	14	7.8	49	2.2	3.1	0.048	59	59
11	3)	**M3AP 160 M**	3GAA 162 101-••C	1460	92.0	92.7	0.81	21.5	7.6	72	2.9	3.4	0.091	107	62
15	3)	**M3AP 160 L**	3GAA 162 102-••C	1460	91.8	92.5	0.82	29	8.1	98	3.3	3.6	0.102	111	62
18.5	3)	**M3AP 180 M**	3GAA 182 101-••C	1470	92.3	92.9	0.84	35	7.0	120	2.9	2.9	0.161	133	62
22	3)	**M3AP 180 L**	3GAA 182 102-••C	1470	93.1	93.9	0.85	40	7.0	143	2.6	2.8	0.225	171	63
30	3)	**M3AP 200 MLB**	3GAA 202 001-••C	1475	93.4	94.0	0.84	55	7.5	194	2.5	2.8	0.34	222	63
37	3)	**M3AP 225 SMA**	3GAA 222 001-••C	1480	93.6	93.8	0.84	68	7.4	239	2.9	3.0	0.37	237	66
45	3)	**M3AP 225 SMB**	3GAA 222 002-••C	1480	94.2	94.4	0.83	83	7.6	291	2.8	3.0	0.42	252	66
55	3)	**M3AP 250 SMA**	3GAA 252 001-••C	1480	94.6	94.9	0.86	98	7.6	355	3.1	3.0	0.72	301	67
75	3)	**M3AP 280 SMA**	3GAA 282 001-••C	1480	94.8	95.3	0.86	132	7.1	486	2.7	3.0	0.88	394	67
90	3)	**M3AP 280 SMB**	3GAA 282 002-••C	1475	95.0	95.3	0.87	157	7.7	583	3.3	3.2	0.95	419	67
4-poles = 1500 r/min				**400 V 50 Hz**									**High-output design**		
18.5	1)	**M3BP 160 LB**	3GBP 162 103-••A	1450	90.5	92.0	0.84	36	6.6	122	2.6	3.0	0.102	135	63
30	1)	**M3BP 180 LB**	3GBP 182 103-••A	1465	92.5	93.3	0.84	56	6.8	196	2.5	2.8	0.225	203	63
37		**M3BP 200 MLB**	3GBP 202 002-••A	1475	93.8	94.2	0.84	68	7.4	240	2.9	2.7	0.34	275	63
55		**M3BP 225 SMC**	3GBP 222 003-••A	1480	94.6	95.0	0.84	100	7.5	356	3.5	3.0	0.49	355	66
75	1)	**M3BP 250 SMB**	3GBP 252 002-••A	1480	94.6	95.1	0.86	132	6.9	486	2.6	3.0	0.88	465	67
110		**M3BP 280 SMC**	3GBP 282 230-••G	1485	95.7	95.7	0.86	195	7.6	707	3.0	3.0	1.85	725	68
250		**M3BP 315 LKA**	3GBP 312 810-••G	1487	96.2	96.2	0.86	442	7.4	1605	2.5	2.9	4.4	1410	78
280		**M3BP 315 LKB**	3GBP 312 820-••G	1487	96.5	96.4	0.87	482	7.6	1798	2.6	3.0	5	1520	78
315		**M3BP 315 LKC**	3GBP 312 830-••G	1488	96.5	96.4	0.86	548	7.8	2022	2.6	3.2	5.5	1600	78
5.5	1) 3)	**M3AP 112 MB**	3GAA 112 002-••C	1425	85.3	86.5	0.83	11.4	6.7	37	2.5	3.2	0.018	36	56
9.2	1) 3)	**M3AP 132 MBA**	3GAA 132 004-••C	1445	87.8	89.2	0.87	17.5	7.2	61	2.7	2.7	0.048	59	59
11	1) 3)	**M3AP 132 MB**	3GAA 132 003-••C	1450	88.6	89.9	0.86	21	7.4	73	2.5	2.7	0.048	63	59
18.5	1) 3)	**M3AP 160 LB**	3GAA 162 103-••C	1450	90.5	92.0	0.84	36	6.6	122	2.6	3.0	0.102	111	63
30	1) 3)	**M3AP 180 LB**	3GAA 182 103-••C	1465	92.5	93.3	0.84	56	6.8	196	2.5	2.8	0.225	170	63
37	3)	**M3AP 200 MLB**	3GAA 202 002-••C	1475	93.8	94.2	0.84	68	7.4	240	2.9	2.7	0.34	222	63
48	1) 3)	**M3AP 200 MLC**	3GAA 202 003-••C	1470	93.6	94.1	0.84	89	8.1	311	4.4	3.2	0.38	287	63
55	3)	**M3AP 225 SMC**	3GAA 222 003-••C	1480	94.6	95.0	0.84	100	7.5	356	3.5	3.0	0.49	287	66
73	1) 3)	**M3AP 225 SMD**	3GAA 222 004-••C	1475	94.2	94.5	0.85	132	8.1	473	3.9	3.2	0.56	312	66
75	3)	**M3AP 250 SMB**	3GAA 252 002-••C	1480	94.6	95.1	0.86	132	6.9	486	2.6	3.0	0.88	357	67
95	1) 3)	**M3AP 250 SMC**	3GAA 252 003-••C	1475	94.5	95.3	0.88	165	7.3	616	2.6	3.1	0.95	386	67

1) Temperature rise class F
2) Efficiency class 2.
3) When ordering, the following variant code has to be added to the product code: 199 = Extreme heavy duty design. Type designation M3AP.

Efficiency classes fixed for ranges 1.1 to 90 kW (available only by 2- and 4-poles).

Process performance cast iron and aluminum motors

Technical data for totally enclosed squirrel cage three phase motors

IP 55, IC 411; Insulation class F, temperature rise class B

Output kW		Type designation	Product code	Speed r/min	Efficiency Full load 100%	Efficiency 3/4 load 75%	Power factor cos φ 100 %	Current I_N A	Current I_s/I_N	Torque T_N Nm	Torque T_s/T_N	Torque T_{max}/T_N	Moment of inertia J=1/4 GD² kgm²	Weight kg	Sound pressure level LP dB(A)
6-poles = 1000 r/min				**400 V 50 Hz**									**Basic design**		
0.18		**M2BA 71 M6 A**	3GBA 073 310-••A	880	57.0	50.4	0.63	0.73	4.0	1.95	1.7	2.4	0.0006	10	42
0.25		**M2BA 71 M6 B**	3GBA 073 320-••A	880	61.5	58.3	0.65	0.91	4.0	2.71	1.7	2.5	0.0007	11	42
0.37		**M2BA 80 M6 A**	3GBA 083 310-••A	920	68.0	63.2	0.65	1.21	5.0	3.84	1.7	2.0	0.0016	17	45
0.55		**M2BA 80 M6 B**	3GBA 083 320-••A	920	70.0	65.1	0.66	1.72	5.0	5.71	1.7	1.8	0.002	18	45
0.75		**M2BA 90 S6 A**	3GBA 093 110-••A	920	74.0	70.2	0.71	2.08	5.0	7.79	2.0	2.3	0.0029	21	48
1.1		**M2BA 90 L6 A**	3GBA 093 510-••A	920	75.0	73.1	0.73	2.9	5.0	11.42	2.0	2.6	0.0038	25	48
1.5		**M2BA 100 L6 A**	3GBA 103 510-••A	930	79.0	75.5	0.73	3.76	5.5	15.4	2.0	2.4	0.01	32	51
2.2		**M2BA 112 M6 A**	3GBA 113 310-••A	940	83.0	81.1	0.73	5.24	5.5	22.35	2.0	2.3	0.0156	40	54
3		**M2BA 132 S6 A**	3GBA 133 110-••A	960	84.5	82.4	0.77	6.67	6.5	29.84	2.0	2.4	0.0312	55	56
4		**M2BA 132 M6 A**	3GBA 133 310-••A	960	85.0	84.1	0.76	8.94	6.5	39.79	2.0	2.9	0.0407	65	56
5.5		**M2BA 132 M6 B**	3GBA 133 320-••A	950	87.0	85.9	0.78	11.7	6.5	55	2.0	3.0	0.0533	75	56
7.5		**M3BP 160 M**	3GBP 163 101-••A	970	89.3	90.4	0.79	15.4	6.6	74	1.9	2.6	0.089	115	59
11		**M3BP 160 L**	3GBP 163 102-••A	970	89.8	90.5	0.78	23	6.9	109	2.1	3.4	0.107	135	59
15		**M3BP 180 L**	3GBP 183 101-••A	970	90.8	91.5	0.78	31	6.8	147	2.0	3.3	0.217	177	59
18.5		**M3BP 200 MLA**	3GBP 203 001-••A	985	91.1	91.7	0.81	36	7.0	180	2.7	2.5	0.37	245	63
22		**M3BP 200 MLB**	3GBP 203 002-••A	980	91.7	92.2	0.81	43	6.8	214	2.9	3.0	0.43	260	63
30		**M3BP 225 SMB**	3GBP 223 001-••A	985	92.8	93.0	0.83	56	7.2	291	3.1	2.9	0.64	320	63
37		**M3BP 250 SMA**	3GBP 253 001-••A	985	93.4	93.7	0.83	69	7.3	358	3.1	2.8	1.16	415	63
45		**M3BP 280 SMA**	3GBP 283 210-••G	990	94.4	94.3	0.84	82	7.0	434	2.5	2.5	1.85	605	66
55		**M3BP 280 SMB**	3GBP 283 220-••G	990	94.6	94.6	0.84	101	7.0	531	2.7	2.6	2.2	645	66
75		**M3BP 315 SMA**	3GBP 313 210-••G	992	95.0	94.7	0.82	141	7.4	722	2.4	2.8	3.2	830	70
90		**M3BP 315 SMB**	3GBP 313 220-••G	992	95.5	95.3	0.84	163	7.5	866	2.4	2.8	4.1	930	70
110		**M3BP 315 SMC**	3GBP 313 230-••G	991	95.6	95.5	0.83	202	7.4	1060	2.5	2.9	4.9	1000	70
132		**M3BP 315 MLA**	3GBP 313 410-••G	991	95.8	95.7	0.83	240	7.5	1272	2.7	3.0	5.8	1150	68
160		**M3BP 355 SMA**	3GBP 353 210-••G	993	96.0	95.8	0.83	293	7.0	1539	2.0	2.6	7.9	1520	75
200		**M3BP 355 SMB**	3GBP 353 220-••G	993	96.2	96.1	0.84	357	7.2	1923	2.2	2.7	9.7	1680	75
250		**M3BP 355 SMC**	3GBP 353 230-••G	993	96.5	96.3	0.83	450	7.4	2404	2.6	2.9	11.3	1820	75
315		**M3BP 355 MLB**	3GBP 353 420-••G	992	96.4	96.3	0.83	570	7.0	3032	2.5	2.7	13.5	2180	75
355		**M3BP 355 LKA**	3GBP 353 810-••G	992	96.6	96.5	0.83	640	7.6	3417	2.7	2.9	15.5	2500	75
400	1)	**M3BP 355 LKB**	3GBP 353 820-••G	992	96.4	96.4	0.83	722	7.2	3851	2.6	2.6	16.5	2600	75
400		**M3BP 400 LA**	3GBP 403 510-••G	993	96.7	96.6	0.82	730	7.1	3847	2.3	2.7	17	2900	76
400		**M3BP 400 LKA**	3GBP 403 810-••G	993	96.7	96.6	0.82	730	7.1	3847	2.3	2.7	17	2900	76
450		**M3BP 400 LKB**	3GBP 403 820-••G	994	96.9	96.7	0.82	818	7.4	4323	2.4	2.8	20.5	3150	76
450		**M3BP 400 LB**	3GBP 403 520-••G	994	96.9	96.7	0.82	818	7.4	4323	2.4	2.8	20.5	3150	76
500		**M3BP 400 LKC**	3GBP 403 830-••G	993	96.9	96.8	0.83	900	7.2	4808	2.5	2.7	22	3300	76
500		**M3BP 400 LC**	3GBP 403 530-••G	993	96.9	96.8	0.83	900	7.2	4808	2.5	2.7	22	3300	76
560		**M3BP 400 LKD**	3GBP 403 840-••G	993	96.9	96.8	0.85	985	7.4	5385	2.4	3.0	24	3400	77
560		**M3BP 400 LD**	3GBP 403 540-••G	993	96.9	96.8	0.85	985	7.4	5385	2.4	3.0	24	3400	77
630		**M3BP 450 LA**	3GBP 453 510-••G	994	97.0	97.0	0.84	1115	6.5	6052	1.1	2.5	31	4150	81
710		**M3BP 450 LB**	3GBP 453 520-••G	995	97.1	97.1	0.85	1240	7.0	6814	1.3	2.5	37	4500	81
800	1)	**M3BP 450 LC**	3GBP 453 530-••G	995	97.1	97.1	0.84	1415	7.2	7678	1.3	2.7	41	4800	81
2.2	2)	**M3AP 112 M**	3GAA 113 001-••C	940	80.5	81.0	0.74	5.4	5.6	22	2.1	2.7	0.015	27	54
3	2)	**M3AP 132 S**	3GAA 133 001-••C	960	84.5	84.8	0.75	6.9	6.5	30	2.1	3.0	0.031	39	61
4	2)	**M3AP 132 MA**	3GAA 133 002-••C	960	85.5	86.1	0.78	8.7	7.1	40	2.6	2.8	0.038	46	61
5.5	2)	**M3AP 132 MB**	3GAA 133 003-••C	955	86.0	87.0	0.78	11.9	6.6	55	2.1	2.8	0.045	54	61
7.5	2)	**M3AP 160 M**	3GAA 163 101-••C	970	89.3	90.4	0.79	15.4	6.6	74	1.9	2.6	0.089	96	59
11	2)	**M3AP 160 L**	3GAA 163 102-••C	970	89.8	90.5	0.78	23	6.9	109	2.1	3.4	0.107	110	59
15	2)	**M3AP 180 L**	3GAA 183 101-••C	970	90.8	91.5	0.78	31	6.8	147	2.0	3.3	0.217	160	59
18.5	2)	**M3AP 200 MLA**	3GAA 203 001-••C	985	91.1	91.7	0.81	36	7.0	180	2.7	2.5	0.37	182	63
22	2)	**M3AP 200 MLB**	3GAA 203 002-••C	980	91.7	92.2	0.81	43	6.8	214	2.9	3.0	0.43	202	63
30	2)	**M3AP 225 SMB**	3GAA 223 001-••C	985	92.8	93.0	0.83	56	7.2	291	3.1	2.9	0.64	247	63
37	2)	**M3AP 250 SMA**	3GAA 253 001-••C	985	93.4	93.7	0.83	69	7.3	358	3.1	2.8	1.16	306	63
45	1) 2)	**M3AP 280 SMA**	3GAA 283 001-••C	985	93.4	93.7	0.84	83	7.2	436	3.2	2.8	1.49	389	63
6-poles = 1000 r/min				**400 V 50 Hz**									**High-output design**		
14	1)	**M3BP 160 LB**	3GBP 163 103-••A	960	89.8	90.1	0.77	29.5	7.0	138	2.5	3.1	0.127	148	62
18.5	1)	**M3BP 180 LB**	3GBP 183 102-••A	965	90.7	91.7	0.80	37	6.1	183	2.1	2.5	0.237	185	59
30	1)	**M3BP 200 MLC**	3GBP 203 003-••A	980	91.9	92.5	0.81	56	7.1	293	3.3	2.9	0.49	275	63
37	1)	**M3BP 225 SMC**	3GBP 223 002-••A	985	92.8	93.4	0.83	69	6.9	360	3.0	2.9	0.75	345	63
45	1)	**M3BP 250 SMB**	3GBP 253 002-••A	985	93.4	93.7	0.84	83	7.2	436	3.2	2.8	1.49	460	63
75		**M3BP 280 SMC**	3GBP 283 230-••G	990	95.1	95.2	0.84	137	7.3	723	2.8	2.7	2.85	725	66
160		**M3BP 315 LKA**	3GBP 313 810-••G	992	95.8	95.7	0.83	293	7.5	1540	2.6	2.8	7.3	1410	74
180		**M3BP 315 LKB**	3GBP 313 820-••G	992	95.9	95.8	0.83	330	7.4	1733	2.6	2.8	8.3	1520	74
200		**M3BP 315 LKC**	3GBP 313 830-••G	989	95.7	95.7	0.84	362	6.8	1931	2.5	2.6	9.2	1600	74
3	1) 2)	**M3AP 112 MB**	3GAA 113 002-••C	935	80.0	81.2	0.76	7.2	5.5	31	2.5	2.7	0.018	33	54
6.3	1) 2)	**M3AP 132 MC**	3GAA 133 004-••C	960	84.9	85.0	0.75	14.5	7.3	63	2.3	3.1	0.049	59	61
14	1) 2)	**M3AP 160 LB**	3GAA 163 103-••C	960	89.8	90.1	0.77	29.5	7.0	138	2.5	3.1	0.127	125	62
18.5	1) 2)	**M3AP 180 LB**	3GAA 183 102-••C	965	90.7	91.7	0.80	37	6.1	183	2.1	2.5	0.237	169	59
30	1) 2)	**M3AP 200 MLC**	3GAA 203 003-••C	980	91.9	92.5	0.81	56	7.1	293	3.3	2.9	0.49	217	63
37	2)	**M3AP 225 SMC**	3GAA 223 002-••C	985	92.8	93.4	0.83	69	6.9	360	3.0	2.9	0.75	274	63
45	1) 2)	**M3AP 250 SMB**	3GAA 253 002-••C	985	93.4	93.7	0.84	83	7.2	436	3.2	2.8	1.49	346	63

1) Temperature rise class F

2) When ordering, the following variant code has to be added to the product code: 199 = Extreme heavy duty design. Type designation M3AP.

Process performance cast iron and aluminum motors

Technical data for totally enclosed squirrel cage three phase motors

IP 55, IC 411; Insulation class F, temperature rise class B

Output kW		Type designation	Product code	Speed r/min	Efficiency Full load 100%	Efficiency 3/4 load 75%	Power factor cos φ 100 %	Current I_N A	Current I_s/I_N	Torque T_N Nm	Torque T_s/T_N	Torque T_{max}/T_N	Moment of inertia J=1/4 GD² kgm²	Weight kg	Sound pressure level LP dB(A)
8-poles = 750 r/min				**400 V 50 Hz**								**Basic design**			
4		**M3BP 160 MA**	3GBP 164 101-••A	715	84.1	84.7	0.69	10	5.1	53	2.1	2.6	0.072	100	59
5.5		**M3BP 160 M**	3GBP 164 102-••A	710	84.7	85.6	0.70	13.4	5.5	74	2.4	2.6	0.091	113	59
7.5		**M3BP 160 L**	3GBP 164 103-••A	715	86.3	87.3	0.70	18.1	5.4	100	2.4	2.7	0.131	126	59
11		**M3BP 180 L**	3GBP 184 101-••A	720	89.6	90.3	0.76	23.5	5.7	146	2.1	2.5	0.224	177	59
15		**M3BP 200 MLA**	3GBP 204 001-••A	740	91.1	91.6	0.82	29	7.5	196	3.0	3.2	0.45	250	60
18.5		**M3BP 225 SMA**	3GBP 224 001-••A	730	91.1	91.6	0.79	37	6.8	242	2.8	3.1	0.61	305	63
22		**M3BP 225 SMB**	3GBP 224 002-••A	730	91.5	92.2	0.77	45	6.4	287	2.4	2.6	0.68	320	63
30		**M3BP 250 SMA**	3GBP 254 001-••A	735	92.8	93.1	0.79	59	7.3	389	2.2	2.6	1.25	415	63
37		**M3BP 280 SMA**	3GBP 284 210-••G	741	93.4	93.3	0.78	74	7.3	477	1.7	3.0	1.85	605	65
45		**M3BP 280 SMB**	3GBP 284 220-••G	741	94.1	93.8	0.78	90	7.6	580	1.8	3.1	2.2	645	65
55		**M3BP 315 SMA**	3GBP 314 210-••G	742	94.1	94.0	0.81	104	7.1	708	1.6	2.7	3.2	830	62
75		**M3BP 315 SMB**	3GBP 314 220-••G	741	94.5	94.4	0.82	141	7.1	968	1.7	2.7	4.1	930	62
90		**M3BP 315 SMC**	3GBP 314 230-••G	741	94.8	94.7	0.82	167	7.4	1161	1.8	2.7	4.9	1000	64
110		**M3BP 315 MLA**	3GBP 314 410-••G	740	95.0	95.0	0.83	203	7.3	1420	1.8	2.7	5.8	1150	72
132		**M3BP 355 SMA**	3GBP 354 210-••G	744	95.7	95.6	0.80	250	7.5	1694	1.5	2.6	7.9	1520	69
160		**M3BP 355 SMB**	3GBP 354 220-••G	744	95.7	95.6	0.80	305	7.6	2054	1.6	2.6	9.7	1680	69
200		**M3BP 355 SMC**	3GBP 354 230-••G	743	95.7	95.6	0.80	378	7.4	2570	1.6	2.6	11.3	1820	69
250		**M3BP 355 MLB**	3GBP 354 420-••G	743	95.9	95.8	0.80	476	7.5	3213	1.6	2.7	13.5	2180	72
315	1)	**M3BP 355 LKB**	3GBP 354 820-••G	742	95.9	95.9	0.80	594	7.9	4054	1.7	2.7	16.5	2600	75
315		**M3BP 400 LA**	3GBP 404 510-••G	744	96.4	96.3	0.81	582	7.0	4043	1.2	2.6	17	2900	71
315		**M3BP 400 LKA**	3GBP 404 810-••G	744	96.4	96.3	0.81	582	7.0	4043	1.2	2.6	17	2900	71
355		**M3BP 400 LB**	3GBP 404 520-••G	743	96.5	96.5	0.83	640	6.8	4563	1.2	2.5	21	3200	71
355		**M3BP 400 LKB**	3GBP 404 820-••G	743	96.5	96.5	0.83	640	6.8	4563	1.2	2.5	21	3200	71
400		**M3BP 400 LC**	3GBP 404 530-••G	744	96.6	96.5	0.82	735	7.4	5134	1.3	2.7	24	3400	71
400		**M3BP 400 LKC**	3GBP 404 830-••G	744	96.6	96.5	0.82	735	7.4	5134	1.3	2.7	24	3400	71
450		**M3BP 450 LA**	3GBP 454 510-••G	744	96.3	96.3	0.82	820	6.2	5776	1.0	2.2	26	3750	82
500		**M3BP 450 LB**	3GBP 454 520-••G	744	96.4	96.4	0.82	910	6.3	6418	1.0	2.3	29	4000	82
560		**M3BP 450 LC**	3GBP 454 530-••G	745	96.6	96.5	0.82	1015	6.5	7178	1.1	2.3	35	4350	82
630	1)	**M3BP 450 LD**	3GBP 454 540-••G	745	96.7	96.6	0.82	1145	6.9	8075	1.2	2.5	41	4800	82
1.5	2)	**M3AP 112 M**	3GAA 114 001-••C	695	74.5	74.6	0.65	4.5	4.1	21	1.9	2.5	0.016	28	52
2.2	2)	**M3AP 132 S**	3GAA 134 001-••C	720	80.5	80.2	0.67	5.9	5.3	29	1.9	2.5	0.038	46	56
3	2)	**M3AP 132 M**	3GAA 134 002-••C	720	82.0	82.0	0.68	7.8	5.5	40	2.4	2.6	0.045	53	56
4	2)	**M3AP 160 MA**	3GAA 164 101-••C	715	84.1	84.7	0.69	10	5.1	53	2.1	2.6	0.072	83	59
5.5	2)	**M3AP 160 M**	3GAA 164 102-••C	710	84.7	85.6	0.70	13.4	5.5	74	2.4	2.6	0.091	96	59
7.5	2)	**M3AP 160 L**	3GAA 164 103-••C	715	86.3	87.3	0.70	18.1	5.4	100	2.4	2.7	0.131	126	59
11	2)	**M3AP 180 L**	3GAA 184 101-••C	720	89.6	90.3	0.76	23.5	5.7	146	2.1	2.5	0.224	156	59
15	2)	**M3AP 200 MLA**	3GAA 204 001-••C	740	91.1	91.6	0.82	29	7.5	196	3.0	3.2	0.45	192	60
18.5	2)	**M3AP 225 SMA**	3GAA 224 001-••C	730	91.1	91.6	0.79	37	6.8	242	2.8	3.1	0.61	232	63
22	2)	**M3AP 225 SMB**	3GAA 224 002-••C	730	91.5	92.2	0.77	45	6.4	287	2.4	2.6	0.68	247	63
30	2)	**M3AP 250 SMA**	3GAA 254 001-••C	735	92.8	93.1	0.79	59	7.3	389	2.2	2.6	1.25	308	63
37	2)	**M3AP 280 SMA**	3GAA 284 001-••C	735	93.0	93.3	0.81	74	7.4	479	2.0	2.6	1.52	389	63
8-poles = 750 r/min				**400 V 50 Hz**								**High-output design**			
8.5	1)	**M3BP 160 LB**	3GBP 164 104-••A	700	85.1	85.7	0.70	21	5.3	114	2.3	2.6	0.131	128	62
15	1)	**M3BP 180 LB**	3GBP 184 102-••A	720	88.7	89.6	0.76	32.5	6.0	199	2.4	2.6	0.24	185	62
18.5		**M3BP 200 MLB**	3GBP 204 002-••A	735	91.4	91.8	0.81	36	7.3	241	2.6	3.1	0.54	275	60
30	1)	**M3BP 225 SMC**	3GBP 224 003-••A	735	90.5	91.3	0.79	64	6.7	391	2.5	3.0	0.8	345	63
37		**M3BP 250 SMB**	3GBP 254 002-••A	735	93.0	93.3	0.81	74	7.4	479	2.0	2.6	1.52	360	63
55		**M3BP 280 SMC**	3GBP 284 230-••G	741	94.4	94.3	0.80	105	7.9	709	1.9	3.1	2.85	725	65
132		**M3BP 315 LKA**	3GBP 314 810-••G	740	95.1	95.2	0.83	243	7.3	1703	1.8	2.6	7.3	1410	74
150		**M3BP 315 LKB**	3GBP 314 820-••G	741	95.3	95.3	0.83	275	7.7	1933	1.9	2.7	8.3	1520	74
160		**M3BP 315 LKC**	3GBP 314 830-••G	740	95.2	95.2	0.83	292	7.7	2065	1.9	2.8	9.2	1600	75
1.9	1) 2)	**M3AP 112 MB**	3GAA 114 002-••C	690	74.0	74.8	0.67	5.6	4.3	26.5	2.0	2.6	0.018	33	52
3.8	1) 2)	**M3AP 132 MB**	3GAA 134 003-••C	710	80.5	80.7	0.69	9.9	5.2	51	2.3	2.6	0.049	59	56
8.5	1) 2)	**M3AP 160 LB**	3GAA 164 104-••C	700	85.1	85.7	0.70	21	5.3	114	2.3	2.6	0.131	126	62
15	1) 2)	**M3AP 180 LB**	3GAA 184 102-••C	720	88.7	89.6	0.76	32.5	6.0	199	2.4	2.6	0.24	164	62
18.5	2)	**M3AP 200 MLB**	3GAA 204 002-••C	735	91.4	91.8	0.81	36	7.3	241	2.6	3.1	0.54	217	60
30	1) 2)	**M3AP 225 SMC**	3GAA 224 003-••C	735	90.5	91.3	0.79	64	6.7	391	2.5	3.0	0.8	277	63
37	2)	**M3AP 250 SMB**	3GAA 254 002-••C	735	93.0	93.3	0.81	74	7.4	479	2.0	2.6	1.52	346	63

1) Temperature rise class F

2) When ordering, the following variant code has to be added to the product code: 199 = Extreme heavy duty design. Type designation M3AP.

Process performance motors - Variant codes

Code	Variant	Cast iron motors 71-132	160-250	280-355	400-450	Aluminum motors 112-132	160-180	200-280
Bearings and Lubrication								
037	Roller bearing at D-end. Transport lock included.	NA	M	M	P	NA	M	M
039	Cold resistant grease.	M	M	P	P	M	M	M
040	Heat resistant grease.	M	S	M	P	S	S	S
041	Bearings regreasable via grease nipples.	NA	S	S	S	M	S	S
043	SPM nipples.	M	S	S	S	M	S	S
057	2RS bearings at both ends.	S	M	NA	NA	M	M	M
Branch standard design								
178	Stainless steel/acid proof bolts.	M	M	M	P	S	S	S
Cooling system								
053	Metal fan cover.	S	S	S	S	S	S	S
068	Metal fan.	M	M	M	P	M	M	M
075	Cooling method IC418 (without fan).	R	M	R	R	R	R	R
183	Separate motor cooling (fan axial, N-end).	M	M	P	P	NA	M	M
Earthing bolt								
067	External earthing bolt.	M	S	S	S	M	M	M
Heating elements								
450	Heating element 100-120 V.	M	M	M	P	M	M	M
451	Heating element 200-240 V.	M	M	M	P	M	M	M
Mounting arrangements								
007	IM 3001 flange mounted, from IM 1001(B5 from B3).	NA	NA	NA	NA	NA	NA	M
008	IM 2101 foot/flange mounted, from IM 1001(B34 from B3).	M	NA	NA	NA	M	NA	NA
009	IM 2001 foot/flange mounted, from IM 1001 (B35 from B3).	M	M	M	P	M	M	M
047	IM 3601 flange mounted, from IM 3001 (B14 from B5).	M	R/NA	NA	NA	M	NA	NA
048	IM 3001 flange mounted, from IM 3601 (B5 from B14).	M	NA	NA	NA	M	NA	NA
066	Modified for non-standard mounting position. (Please specify IMxxxx.) Must be ordered for all mounting arrangements excluding IM B3 (1001) and B5 (3001).	NA	M	M	P	M	M	M
Painting								
114	Special paint colour, standard grade.	M	M	M	P	M	M	M
179	Special paint specification.	R	R	R	R	R	R	R
Protection								
005	Protective roof, vertical motor, shaft down. Vertically mounted motors with shaft extension downwards.	M	M	M	P	M	M	M
072	Radial seal at D-end.	M	M	M	P	M	M	M
158	Degree of protection IP65 or IP65X. Dust proof version.	M	M	M	P	M	M	M
211	Weather protected, IP xx W.	P	P	P	P	M	M	M
403	Degree of protection IP56.	M	M	M	P	M	M	M
Rating & instruction plates								
002	Restamping voltage, frequency and output, continuous duty. All data to be specified.	M	M	M	P	M	M	M
003	Individual serial number.	M	S	S	S	M	M	M
098	Stainless rating plate.	S	S	S	S	S	S	S
138	Mounting of additional identification plate, aluminum.	M	NA	NA	NA	M	M	M
Stator winding temperature sensors								
122	Bimetal detectors, break type (NCC), (3 in series), 150°C.	M	M	P	P	M	M	M
436	PTC - thermistors (3 in series), 150 °C.	S	S	S	S	M	S	S
439	PTC - thermistors (2x3 in series), 150 °C.	M	M	M	P	M	M	M
445	PT100 (1 per phase).	NA/M	M	M	P	M	M	M
Terminal box								
021	Terminal box left-hand side, seen from D-end.	NA/P	P	P	P/NA	NA	NA	P
180	Terminal box on right-hand side, seen from D-end.	NA/P	P	P	P/NA	NA	NA	P
230	Standard gable glands.	M	M	S	S	M	M	M
731	Two standard cable glands.	M	M	S	S	M	M	M
743	Painted steel flange for cable glands.	NA	M	M	P	NA	NA	NA
Testing								
145	Type test report from test of identical motor. 400V 50 Hz.	M	M	M	P	M	M	M
148	Routine test report.	M	M	M	P	M	M	M
Variable speed drives								
701	Insulated bearing at N-end.	NA	M/R	M	P	NA	NA	M
704	EMC cable gland.	NA	M	M	P	M	M	M

Certain variant codes cannot be used together.

Note! This list is an extract of a wide range of possible modifications. Please see the General Purpose or Process Performance Motor Catalogue for a complete list of variant codes.

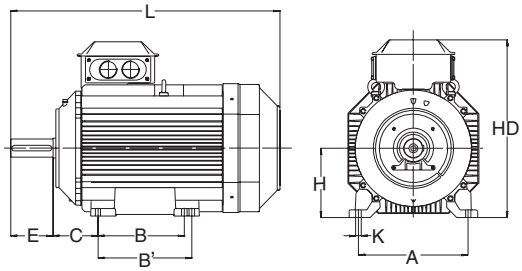
S = Included as standard.
M = On modification of a stocked motor, or on new manufacture,the number per order may be limited.
P = New manufacture only.
R = On request.
NA = Not applicable.

Motors in brief, Basic design

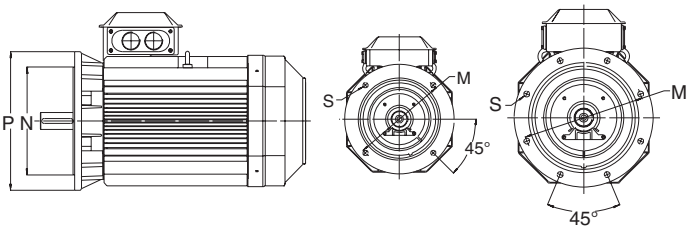
Size		71	80	90	100	112	132	160	180	200	225	250	280	315	355	400	450
Process performance motors, cast iron																	
Bearings	D-end, 2 pole	6202	6204	6205	6206	6207	6208	6309/C3	6310/C3	6312/C3	6313/C3	6315/C3	6316/C3	6316/C3	6316M/C3	6317M/C3	6317M/C3
	D-end, 4-8 pole	2RS C3	2RS C3	2RS C3	2RS C3	2RS C3	2RS C3						6316/C3	6319/C3	6322/C3	6324/C3	6326M/C3
	N-end, 2 pole	6202	6204	6205	6206	6206	6207	6309/C3	6309/C3	6310/C3	6312/C3	6313/C3	6316/C3	6316/C3	6316M/C3	6317M/C3	6317M/C3
	N-end, 4-8 pole	2RS C3	2RS C3	2RS C3	2RS C3	2RS C3	2RS C3						6316/C3	6316/C3	6316/ C3	6319/C3	6322/C3
Axially-locked bearings	Inner bearing cover	As standard, locked at D-end.															
Lubrication		Greased for life						Regreasable bearings, regreasing nipples M6x1.					Regreasable bearings, regreasing nipples M10x1.				
SPM-nipples		Optional						As standard.									
Connections	Cable entries	2xM16	2xM25		2xM32			2xM40		2xM63			2xM63+2xM20		2xØ60/80		
	Screw	M4			M5			M6		M10			M12				
	Terminal box	6 terminals for connection with cable lugs (not included).															
	Cable glands	Optional						Cable flanges as standard, cable glands as option.					Cable glands included as standard.				
Winding protection		3 PTC thermistors as standard, 150 °C.											3 PTC thermistors as standard, 155 °C.				
Drain holes		Optional.						Standard.									
Process performance motors, aluminum																	
Bearings	D-end					6306-2Z/C3	6308-2Z/C3	6309/C3	6310/C3	6312/C3	6313/C3	6315/C3	6316/C3 [1]				
	N-end					6206-2Z/C3	6208-2Z/C3	6309/C3	6309/C3	6310/C3	6312/C3	6313/C3	6313/C3	[1] 6315/C3 for 2-pole motors.			
Axially-locked bearings	Inner bearing cover					D-end [1]		D-end									
		[1] Foot motor. A spring washer at the N-end presses the motor towards the D-end. Flange motor. Inner bearing cover and spring-washer at the N-end.															
Lubrication						Permanently lubricated shielded bearings.		Valve lubrication.									
SPM-nipples						Optional.		As standard.									
Connections	Cable entries					2x(M25+M20)		2x(2xM40+M16)		1x(2xM40+M16)			1x2xM63+M16				
	Screw					M5		M6		M10							
	Terminal box					6 terminals for connection with cable lugs (not included).											
Winding protection						Optional.		PTC thermistors, 150°C, 3 in series.									
Drain holes						Standard at both ends.											

Dimension drawings

Foot-mounted motor IM 1001, B3



Flange-mounted motor IM 3001, B5



Motor size	IM 1001, IM B3 and IM 3001, IM B5										IM 1001, IM B3							IM 3001, IM B5			
	D poles		GA poles		F poles		E poles		L max poles		A	B	B'	C	HD	K	H	M	N	P	S
	2	4-8	2	4-8	2	4-8	2	4-8	2	4-8											
Aluminum frame, general purpose and process performance motors																					
M2VA 56	9	9	10.2	10.2	3	3	20	20	197	197	90	71	–	36	159	5.8	56	100	80	120	7
63	11	11	12.5	12.5	4	4	23	23	205	205	100	80	–	40	171	7	63	115	95	140	10
71	14	14	16	16	5	5	30	30	238	238	112	90	–	45	176	7	71	130	110	160	10
80	19	19	21.5	21.5	6	6	40	40	265	265	125	100	–	50	190	10	80	165	130	200	12
M3AA90 S	24	24	27	27	8	8	50	50	282	282	140	100	–	56	217	10	90	165	130	200	12
90 L	24	24	27	27	8	8	50	50	307	307	140	125	–	56	217	10	90	165	130	200	12
100	28	28	31	31	8	8	60	60	349	349	160	140	–	63	237	12	100	215	180	250	15
M2AA112	28	28	31	31	8	8	60	60	361 [2]	361 [2]	190	140	–	70	258	12	112	215	180	250	14.5
132	38	38	41	41	10	10	80	80	447 [3]	447 [3]	216	140	178 [1]	89	295.5	12	132	265	230	300	14.5
160	42	42	45	45	12	12	110	110	602.5	602.5	254	210	254	108	370	15	160	300	250	350	19
180 M	48	48	51.5	51.5	14	14	110	110	602.5	602.5	279	241	279	121	390	15	180	300	250	350	19
180 L	48	48	51.5	51.5	14	14	110	110	643.5	643.5	279	241	279	121	390	15	180	300	250	350	19
200 LA	55	55	59	59	16	16	110	110	711.5	711.5	318	267	305	133	425	18	200	350	300	400	19
200 L 2-4	55	55	59	59	16	16	110	110	732	732	318	267	305	133	425	18	200	350	300	400	19
225 M	55	65	59	64	16	18	110	140	773	843	356	286 [1]	311	149	525.5	18	225	400	350	450	19
225 S		60		64		18		140		803	356	286	311 [1]	149	525.5	18	225	400	350	450	19
250 M	60	65	64	69	18	18	140	140	866	866	406	311 [1]	349	168	571	22	250	500	450	550	19
M3AA/M3AP																					
112M 6-8		28		31		8		60		361 [2]	190	140	–	70	258	12	112	215	180	250	14.5
112 M 2-4, MB	28	28	31	31	8	8	60	60	388 [2]	388 [2]	190	140	–	70	258	12	112	215	180	250	14.5
132 SA,S, MA,MB 6,M8	38	38	41	41	10	10	80	80	447 [3]	447 [3]	216	140	178 [1]	89	295.5	12	132	265	230	300	14.5
132 all exc. above	38	38	41	41	10	10	80	80	481.5[3]	481.5[2]	216	140 [1]	178	89	295.5	12	132	265	230	300	14.5
160 M/MA 2-8, L 2-6, LB 2-4	42	42	45	45	12	12	110	110	602.5	602.5	254	210	254	108	370	15	160	300	250	350	19
160 L 8, LB 6-8	42	42	45	45	12	12	110	110	643.5	643.5	254	210	254	108	370	15	160	300	250	350	19
180 M 2-4,L 6-8,LB 2	48	48	51.5	51.5	14	14	110	110	680	680	279	241	279	121	405	15	180	300	250	350	19
180 L 4, LB 4-8	48	48	51.5	51.5	14	14	110	110	700.5	700.5	279	241	279	121	405	15	180	300	250	350	19
200 MLD-2,-C 4	55	55	59	59	16	16	110	110	814	814	318	267	305	133	533	18	200	350	300	400	19
200 all exc. above	55	55	59	59	16	16	110	110	774	774	318	267	305	133	533	18	200	350	300	400	19
225 SMB, -C	55	55	59	59	16	16	110	110	836	836	356	286	311	149	578	18	225	400	350	450	19
225 SMA,-B,-C	60	60	64	64	18	18	140	140	866	891	356	286	311	149	578	18	225	400	350	450	19
225 SMD	55	60	59	64	16	18	110	140	861	891	356	286	311	149	578	18	225	400	350	450	19
250 SMA,-B	60	65	64	69	18	18	140	140	875	875	406	311	349	168	626	22	250	500	450	550	19
250 SMC	60	65	64	69	18	18	140	140	900	900	406	311	349	168	626	22	250	500	450	550	19
280 SMA	65	75	69	79.5	18	20	140	140	875	875	457	368	419	190	656	24	280	500	450	550	19
280 SMB	65	75	69	79.5	18	20	140	140	900	900	457	368	419	190	656	24	280	500	450	550	19
Steel frame, general purpose																					
M2CA280 SA	65	75	69	79.5	18	20	140	140	1060	990	457	368	–	190	730	24	280	500	450	550	18
280 SMA	65	75	69	79.5	18	20	140	140	1060	1060	457	368	419	190	730	24	280	500	450	550	18
280 MB	65	75	69	79.5	18	20	140	140	1120	1120	457	419	–	190	730	24	280	500	450	550	18
280 MC	65	75	69	79.5	18	20	140	140	1255	1255	457	419	–	190	730	24	280	500	450	550	18
280 MD	65	75	69	79.5	18	20	140	140	1255	1255	457	419	–	190	730	24	280	500	450	550	18
315 SA	65	80	69	85	18	22	140	170	1095	1125	508	406	–	216	820	28	315	600	550	660	23
315 SMA	65	80	69	85	18	22	140	170	1195	1125	508	406	457	216	820	28	315	600	550	660	23
315 MB	65	80	69	85	18	22	140	170	1195	1225	508	457	–	216	820	28	315	600	550	660	23
315 LA	65	90	69	95	18	25	140	170	1265	1295	508	508	–	216	820	28	315	600	550	660	23
315 LB	65	90	69	95	18	25	140	170	1545	1575	508	508	–	216	820	28	315	600	550	660	23
315 LC	65	90	69	95	18	25	140	170	1545	1575	508	508	–	216	848	28	315	600	550	660	23
315 LC; 6 pole	–	90	–	95	–	25	–	170	–	1575	508	508	–	216	820	28	315	600	550	660	23
355 SA, SB	70	100	74.5	106	20	28	140	210	1317	1387	610	500	–	254	915	28	355	740	680	800	23
355 MA, MB	70	100	74.5	106	20	28	140	210	1377	1447	610	560	–	254	915	28	355	740	680	800	23
355 LA, LB	70	100	74.5	106	20	28	140	210	1457	1527	610	630	–	254	915	28	355	740	680	800	23
355 LKD	–	100	–	106	–	28	–	210	–	1667	610	630	710	254	915	28	355	740	680	800	23
400 MLA,MLB	70	100	74.5	106	20	28	140	210	1628	1698	686	630	710	280	1000	35	400	740	680	800	23
400 LKA,LKB	80	100	85	106	22	28	170	210	1798	1838	686	710	800	280	1000	35	400	740	680	800	23

IM 3601, IM B14

Motor size	M	N	P	S
56	65	50	80	M5
63	75	60	90	M5
71	85	70	105	M6
80	100	80	120	M6
90	115	95	140	M8
100	130	110	160	M8
112	130	110	160	M8
132	165	130	200	M10

Tolerances:

A, B	± 0,8
D, DA	ISO k6 < Ø 50mm
	ISO m6 > Ø 50mm
F, FA	ISO h9
H	+0 -0.5
N	ISO j6
C, CA	± 0.8

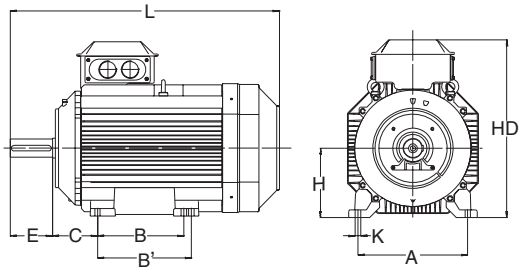
[1] Not according to IEC.
[2] For variant code 053 increased by 7.5 mm.
[3] For variant code 053 increased by 5.5 mm.

Above table gives the main dimensions in mm.

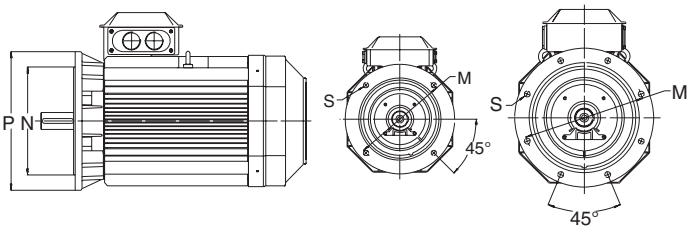
For detailed drawings please check our web-site 'www.abb.com/motors&drives' or contact ABB.

Dimension drawings

Foot-mounted motor IM 1001, B3



Flange-mounted motor IM 3001, B5



Motor size	IM 1001, IM B3 and IM 3001, IM B5										IM 1001, IM B3							IM 3001, IM B5			
	D poles		GA poles		F poles		E poles		L max poles		A	B	B'	C	HD	K	H	M	N	P	S
	2	4-8	2	4-8	2	4-8	2	4-8	2	4-8											
Cast iron frame, general purpose motors																					
M2QA 71 M	14	14	16	16	5	5	30	30	255	255	112	90	–	45	200	7	71	130	110	160	10
80 M	19	19	21.5	21.5	6	6	40	40	285	285	125	100	–	50	225	10	80	165	130	200	12
90 S	24	24	27	27	8	8	50	50	310	310	140	100	–	56	240	10	90	165	130	200	12
90 L	24	24	27	27	8	8	50	50	335	335	140	125	–	56	240	10	90	165	130	200	12
100 L	28	28	31	31	8	8	60	60	380	380	160	140	–	63	275	12	100	215	180	250	15
112 M	28	28	31	31	8	8	60	60	380	380	190	140	–	70	290	12	112	215	180	250	15
132 S	38	38	41	41	10	10	80	80	465	465	216	140	–	89	335	12	132	265	230	300	15
132 M	38	38	41	41	10	10	80	80	505	505	216	178	–	89	335	12	132	265	230	300	15
160 M	42	42	55	55	12	12	110	110	600	600	254	210	–	108	415	15	160	300	250	350	19
160 L	42	42	55	55	12	12	110	110	645	645	254	254	–	108	415	15	160	300	250	350	19
180 M	48	48	51.5	51.5	14	14	110	110	670	670	279	241	–	121	450	15	180	300	250	350	19
180 L	–	48	–	51.5	–	14	–	110	–	710	279	279	–	121	450	15	180	300	250	350	19
200 L	55	55	59	59	16	16	110	110	770	770	318	305	–	133	510	19	200	350	300	400	19
225 S	–	60	–	64	–	18	–	140	–	820	356	286	–	149	560	19	225	400	350	450	19
225 M	55	60	59	64	16	18	110	140	815	840	356	311	–	149	560	19	225	400	350	450	19
250 M	60	55	64	69	18	18	140	140	930	930	406	349	–	168	645	24	250	500	450	550	18
M2BAT 280 SM_	65	75	69	79.5	18	20	140	140	1088	1088	457	368	419	190	745	24	280	500	450	550	18
315 SM_	65	80	69	85	18	22	140	170	1218	1248	508	406	457	216	840	30	315	600	550	660	23
315 ML_	65	90	69	95	18	25	140	170	1269	1299	508	457	508	216	840	30	315	600	550	660	23
355 S	70	100	74.5	106	20	28	140	210	1344	1414	610	500	–	254	955	35	355	740	680	800	23
Cast iron frame, process performance motors																					
M2BA 71	14	14	16	16	5	5	30	30	255	255	112	90	–	45	190	7	71	130	110	160	10
80	19	19	21.5	21.5	6	6	40	40	285	285	125	100	–	50	225	10	80	165	130	200	12
90 S	24	24	27	27	8	8	50	50	310	310	140	100	–	56	240	10	90	165	130	200	12
90 L	24	24	27	27	8	8	50	50	335	335	140	125	–	56	240	10	90	165	130	200	12
100	28	28	31	31	8	8	60	60	380	380	160	140	–	63	275	12	100	215	180	250	15
112	28	28	31	31	8	8	60	60	380	380	190	140	–	70	290	12	100	215	180	250	15
132 S	38	38	41	41	10	10	80	80	465	465	216	140	–	89	335	12	132	265	230	300	15
132 M	38	38	41	41	10	10	80	80	505	505	216	178	–	89	335	12	132	265	230	300	15
M3BP 160 M/MA,L2-6,LB2	42	42	45	45	12	12	110	110	602.5	602.5	254	210	254	108	382	14.5	160	300	250	350	19
160 L8, LB6-8	42	42	45	45	12	12	110	110	643.5	643.5	254	210	254	108	382	14.5	160	300	250	350	19
180 M, L6-8, LB2	48	48	51.5	51.5	14	14	110	110	680	680	279	241	279	121	422	14.5	180	300	250	350	19
180 L4, LB4-8	48	48	51.5	51.5	14	14	110	110	700.5	700.5	279	241	279	121	422	14.5	180	300	250	350	19
200 ML	55	55	59	59	16	16	110	110	774	774	318	267	305	133	506	18.5	200	350	300	400	19
225 SM	55	60	59	64	16	18	110	140	836	866	356	286	311	149	552	18.5	225	400	350	450	19
250 SM	60	65	64	69	18	18	140	140	845	875	406	311	349	168	605	24	250	500	450	550	19
280 SM_	65	75	69	79.5	18	20	140	140	1088	1088	457	368	419	190	762	24	280	500	450	550	18
315 SM_	65	80	69	85	18	22	140	170	1174	1204	508	406	457	216	852	28	315	600	550	660	23
315 ML_	65	90	69	95	18	25	140	170	1285	1315	508	457	508	216	852	28	315	600	550	660	23
315 LK_	65	95	69	95	18	25	140	170	1491	1521	508	508	560	216	880	28	315	600	550	660	23
355 SM_	70	100	74.5	106	20	28	140	210	1409	1479	610	500	560	254	958	35	355	740	680	800	23
355 ML_	70	100	74.5	106	20	28	140	210	1514	1584	610	560	630	254	958	35	355	740	680	800	23
355 LK_	70	100	74.5	106	20	28	140	210	1764	1834	610	630	710	254	958	35	355	740	680	800	23
400 L_	80	110	85	126	22	28	170	210	1851	1891	710	900	1000	224	1045	35	400	940	880	1000	28
400 LK_	80	100	85	106	22	28	170	210	1851	1891	686	710	800	280	1045	35	400	740	680	800	24
450	80	120	85	127	22	32	170	210	2147	2187	800	1000	1120	250	1169	42	450	1080	1000	1150	28

IM 3601, IM B14

Motor size	M	N	P	S
71	85	70	105	M6
80	100	80	120	M6
90	115	95	140	M8
100	130	110	160	M8
112	130	130	160	M8

Tolerances:

A, B	± 0,8
D, DA	ISO k6 < Ø 50mm ISO m6 > Ø 50mm
F, FA	ISO h9
H	+0 -0.5
N	ISO j6
C, CA	± 0.8

Above table gives the main dimensions in mm.

For detailed drawings please check our web-site 'www.abb.com/motors&drives' or contact ABB.

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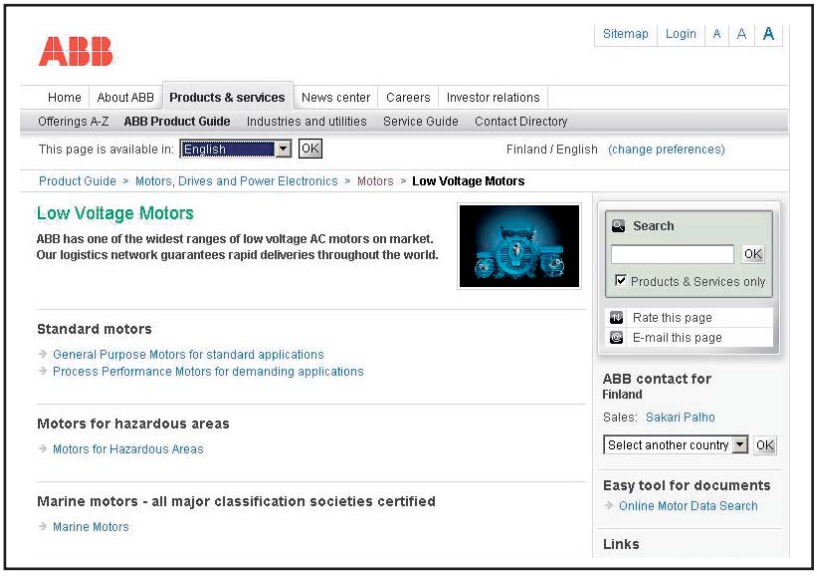

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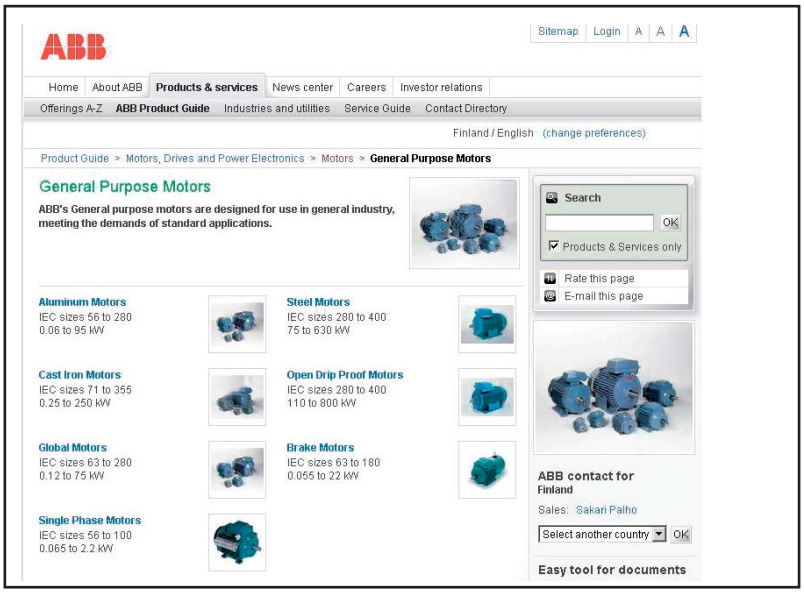

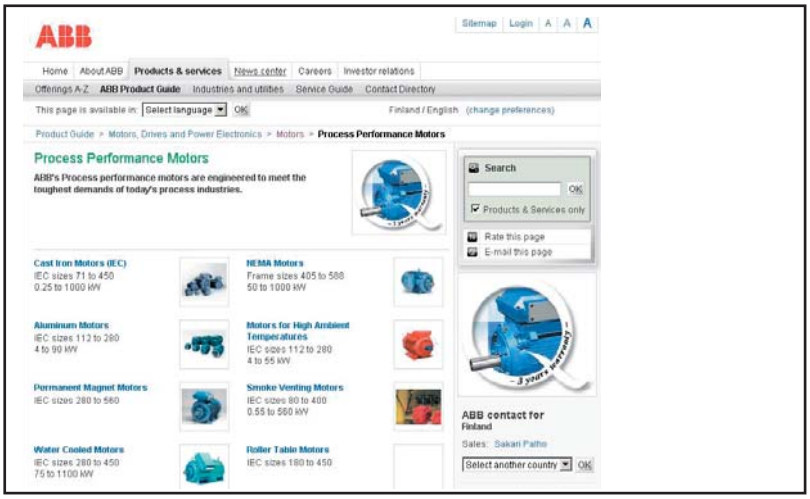

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